



Shell Foundation |



Unlocking Climate Finance to Accelerate Energy Access in Africa

April 22, 2021

Prepared by:



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OFF GRID ADVISORS



→ Commissioned by **Shell Foundation & Rockefeller Foundation**

Shell Foundation | 



→ Produced by **Catalyst Off Grid Advisors**



→ Backup genset research produced in association with the **Schatz Energy Research Center**



→ Inputs from a **Technical Working Group**, comprised of representatives from the following organizations



IKEA Foundation



→ Backup genset research was based on previous work commissioned by the **International Finance Corporation**



→ The Report is endorsed by key **DRE industry associations**



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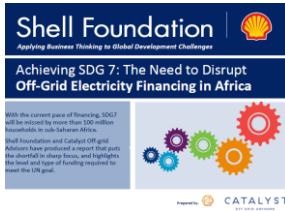
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Call to Action

Summary

This research demonstrates the business opportunity to unlock billions in climate finance and deliver on multiple SDG goals



- Our [2018 research](#) demonstrated the financing opportunity to achieve universal household electrification in Africa (SDG 7) via off-grid solutions
- This research will show **off-grid solar's social dividends**, which cut across numerous SDGs¹
- Alongside the **climate dividends** attributable to low-carbon SDG 7 scenarios
- It forecasts the **climate finance opportunity** associated with these low-carbon SDG 7 scenarios
- **Illustrating the multi-billion-dollar** climate finance opportunity associated with the low-carbon scenarios

¹ Forthcoming research will illustrate how DREs impact SDGs 1, 5, 8, and 10.

Research focus: predictive modeling illustrates Africa's low-carbon SDG 7 scenarios and the impact they will have on SDG 13

- **SDG 7:** Ensure access to affordable, reliable, sustainable and modern energy for all
- **SDG 13:** Take urgent action to combat climate change and its impacts

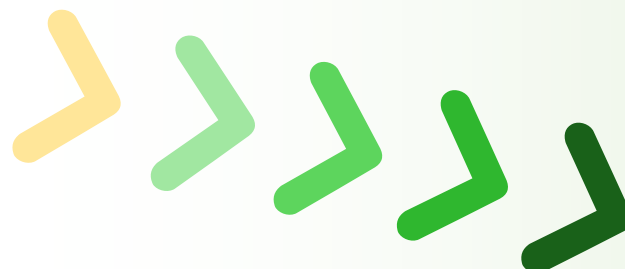
**Universal access to
modern, reliable electricity
by 2030**



**High carbon scenario
toward electricity access**



Africa GHG emissions from access



**2°C or less low carbon scenario
toward electricity access**



**Reduce 41% of
global emissions**



Climate Imperative

Africa's key energy trends and their climate impacts illustrate the scope of the SDG 7 and SDG 13 challenges



Electricity access

- >> **70%** of African households are unelectrified, meaning **200M** need to be connected to reach SDG 7
- >> Africa is falling behind the rest of the world on electricity access, hosting **69%** of the world's unelectrified households
- >> On top of that, Africa remains heavily dependent on fossil fuels, which accounts for **68%** of electricity generation



Unreliable grid access

- >> **Two-thirds** of African grids are considered unreliable, with enterprises experiencing an average of **10%** downtime, and **8%** revenue losses
- >> As a consequence, there are **~ 7 million** backup gensets deployed on the continent, equivalent to **120** coal power stations
- >> These gensets consume **US\$13 billion/year** of fossil fuels

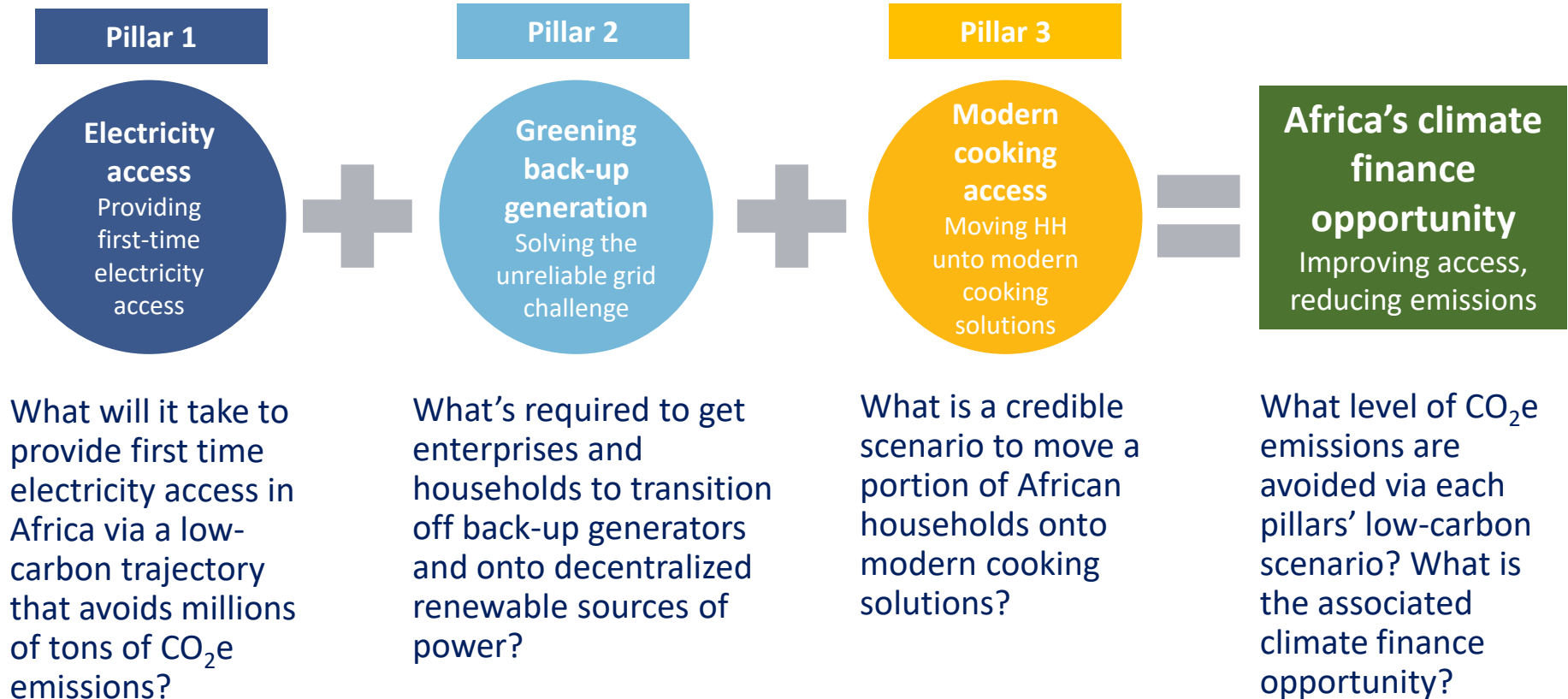


Clean cooking

- >> **82%** of Africa's population (890 million people) use solid fuels for primary cooking needs
- >> **600,000** Africans are killed annually from household air pollution, making it the 2nd largest health risk on the continent
- >> **600 Mt** CO₂ comes from solid cooking in Africa alone

Low-carbon scenarios accelerate Africa's achievement of SDG 7 and SDG 13 via 3 pillars

Predictive modeling forecasts three scenarios for each thematic pillar: business-as-usual, high-carbon, and low-carbon, shows the avoided emissions between the latter two, and then provides the investment costs associated with the low-carbon scenario



These scenarios avoid 563-626 million tons of CO₂, deliver significant SDG 7 impacts, and unlock a substantial climate finance opportunity

Emissions from plausible low-carbon scenarios are benchmarked against equally plausible high-carbon counterfactuals; the difference between the two constitutes the avoided emissions



Summarizing each pillar's social and climate impacts, and the associated climate finance needs of each low-carbon scenario

	BAU ¹ @ 2030	 impact	 technology mix	 avoided emissions	 financing opportunity
Pillar 1: Electricity Access					
Tier 1	• 69% household (HH) access rate	• 100% HH access • 186 million new HH connections	• 43% Off-grid • 2% Mini-grid • 55% Grid	268.1 Mt CO ₂	\$20.7 billion
MEM			• 35% Off-grid • 10% Mini-grid • 55% Grid	205 Mt CO ₂	\$67.1 billion
Pillar 2: Greening Back-up Generators					
Back-up gensets	• Gensets that reach end-life are replaced w/fossil-gensets	• 9.2 million back-up gensets displaced w/ DRE	• Each year, 50% of end-life gensets replaced w/DREs	226.4 Mt CO ₂	\$134.4 billion
Pillar 3: Improved Cooking					
Clean Cooking	• 39 million HHs continue to cook with charcoal	• 39 million HHs transition to cook with modern fuels	• 60% LPG • 22% electricity • 11% ethanol • 7% pellets	131.7 Mt CO ₂	\$7.5 billion

Africa's low-carbon access scenarios: huge impact, significant avoided CO₂ emissions, large climate finance opportunity



A **low-carbon scenario** contributes massively toward **universal access** and **improved cooking**



132 million

new connections from **off-grid** technologies delivered

9.2 million

gensets used by enterprises and households **displaced**

39 million

new households would **cook with modern fuels**



A **low-carbon scenario** benchmarked against a high-carbon scenario **yields**



Up to **626 million**

tons of **avoided CO₂e emissions** over the next decade



A **low-carbon scenario** requires substantial volumes of **new capital**



US\$200+ billion

climate finance opportunity

And creates an imperative to rally climate-first investors who can mobilize commercial and concessional capital to deliver on SDG7

Climate finance must be mobilized at scale to support energy access; several types of institutions need to be brought to the table to change this

Public Organizations

Bilateral DFIs

Single country owns institution and directs finance flows

Multilateral Regional DFIs

Multiple shareholder countries and directs finance flows

Climate Funds

National or multinational climate funds

Private Organizations

Corporations

Project developers and corporate actors

Family Offices

Philanthropic and/or commercial financing

Private Equity / VC

Entities that invest in private companies, or that engage in buyouts of public companies

Investment Banks / Institutional Investors

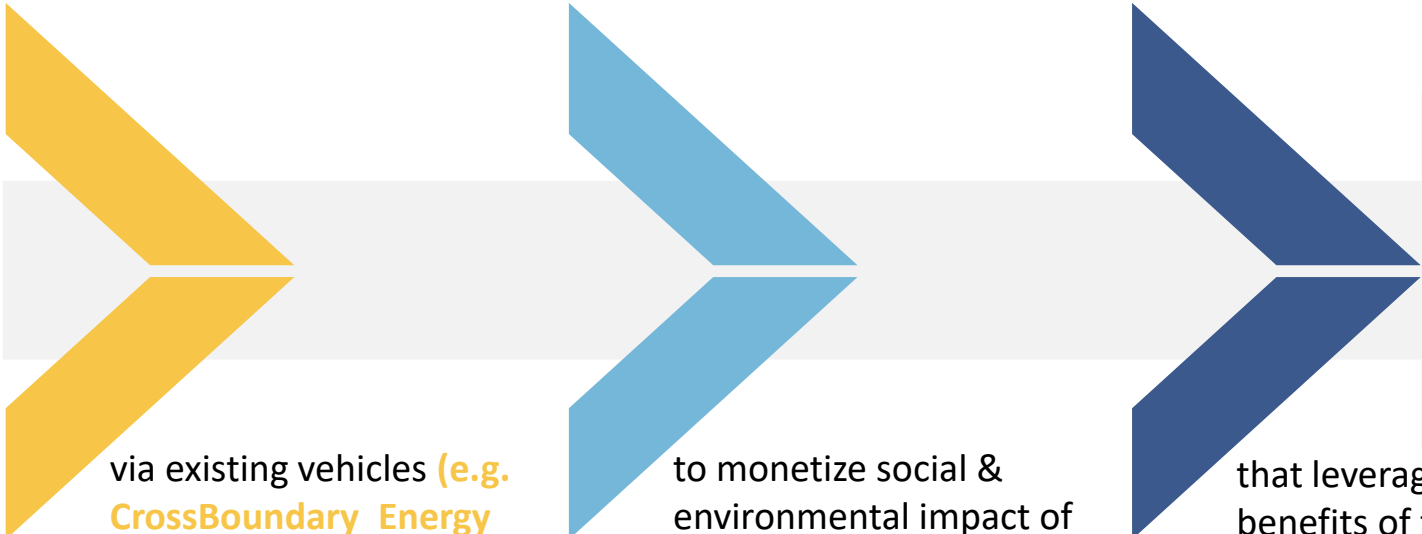
Company or organization that invests money on behalf of others or providers of debt and equity

Three calls to action to climate-first investors to help catalyze the SDG7- climate finance nexus

1 Finance existing energy access enterprises

2 Support new, innovative mechanisms

3 Help define the next wave of investment opportunities



via existing vehicles (e.g. **CrossBoundary Energy Access Facility, Energy Access Ventures, Facility for Energy Inclusion**) to quickly scale impact and get Africa on the low-carbon SDG7 trajectory.

to monetize social & environmental impact of DRE enterprises, such as **Universal Energy Facility, Decentralized Renewable Energy Credits, and Digital Carbon Credits.**

that leverage the co-benefits of the SDG7-SDG13 nexus & **roll out new solutions to unlock climate funding for the DRE sector.**

**\$200+
Billion**

*Climate Finance
Catalysed*

Linking SDG 7 to SDG 13

“Light is a human right”¹: electrification is a crucial social imperative

SUSTAINABLE DEVELOPMENT GOAL 7

Ensure access to affordable, reliable, sustainable and modern energy for all



➤ Yet, **789 million people** still lack **access** to modern electricity in the world



and **SDG 7** is integrally linked other global social imperatives



At the same time, combatting climate change is a top global priority

SUSTAINABLE DEVELOPMENT GOAL 13

Take urgent action to combat climate change and its impacts*



Economic, Social Costs

- **\$3 trillion:** disaster-related economic loss
- **1.3 million lives** claimed by climate-related disasters

Limiting Global Warming

- **2°C or less:** global warming must not exceed this level
- **Decarbonizing the electricity sector** a top priority
 - Via mitigation measures

Climate Finance

- **\$540 billion:** total climate finance flows in 2018
- **\$1.6-3.8T** in climate financing would be needed annually to reach a 1.5° C scenario

Via **Mitigation** to address causes of climate change



Via **Adaptation** to address impacts of climate change

Electricity consumption is a significant contributor to global CO₂ emissions, which leads to the energy-climate nexus

2018 GHG emissions related to electricity and heat

Global Trends

- Accounted for **42% of total energy-related CO₂ emissions**
- Coal combustion** responsible for **44% of global energy-related CO₂ emissions**



Africa Trends

- If **Africa¹**, India emissions reach EU levels (~7/tCO₂), global emissions will increase by 33% (**13/GtCO₂**)

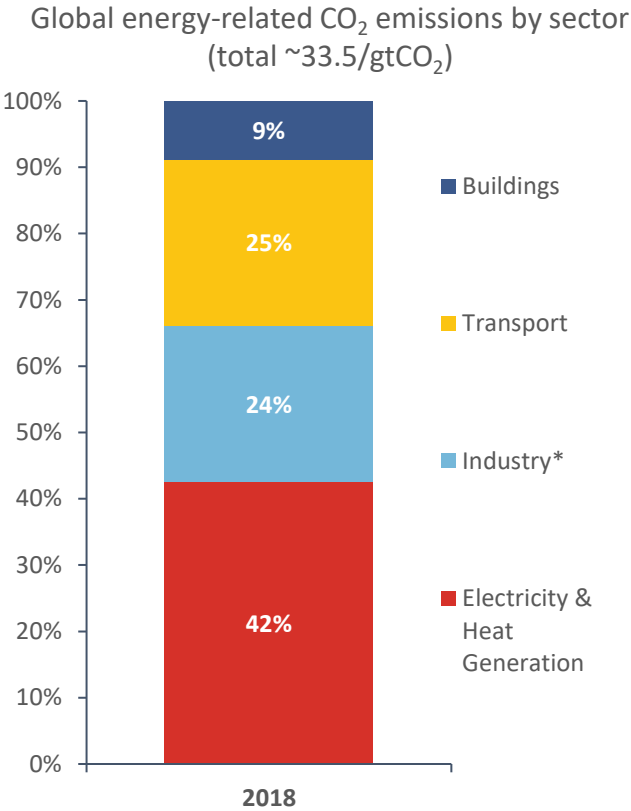


Paris Agreement

- Limiting global warming to **under 2°C** requires **deep decarbonization in electricity sector**



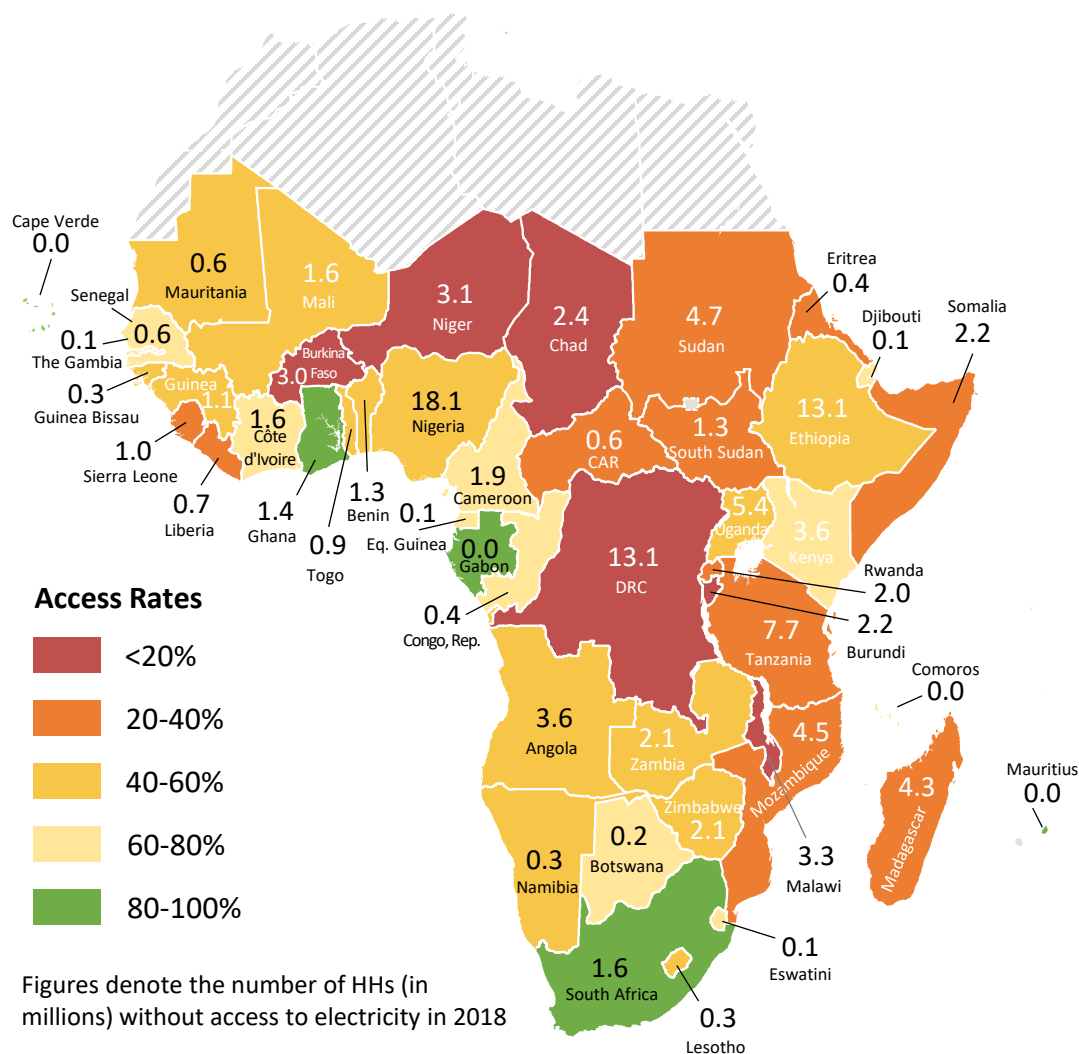
PARIS2015
UN CLIMATE CHANGE CONFERENCE
COP21·CMP11



Source: IEA CO₂ Emissions from Fuel Combustion 2020 (publicly available); ¹ Industry figures also include emissions resulting from energy industry use of energy for own operations

Why Africa's electricity access deficit matters

Africa¹ faces a massive electricity access deficit



~70 percent

of households **that are unelectrified** are found in sub-Saharan Africa

~110 million

African households lack access to modern electricity services in 2020

200 million

households will need to be connected by 2030 to meet **SDG 7** given current demographic trends

These 110 million unelectrified households make do with stopgap solutions for basic lighting

		Fires	Health (smoke, fumes)	Environ. Impact	Cost	Lighting quality
Wood						
Candles						
Kerosene						
Battery operated flashlights						
Mobile phone Flashlights						
Low quality solar lanterns						

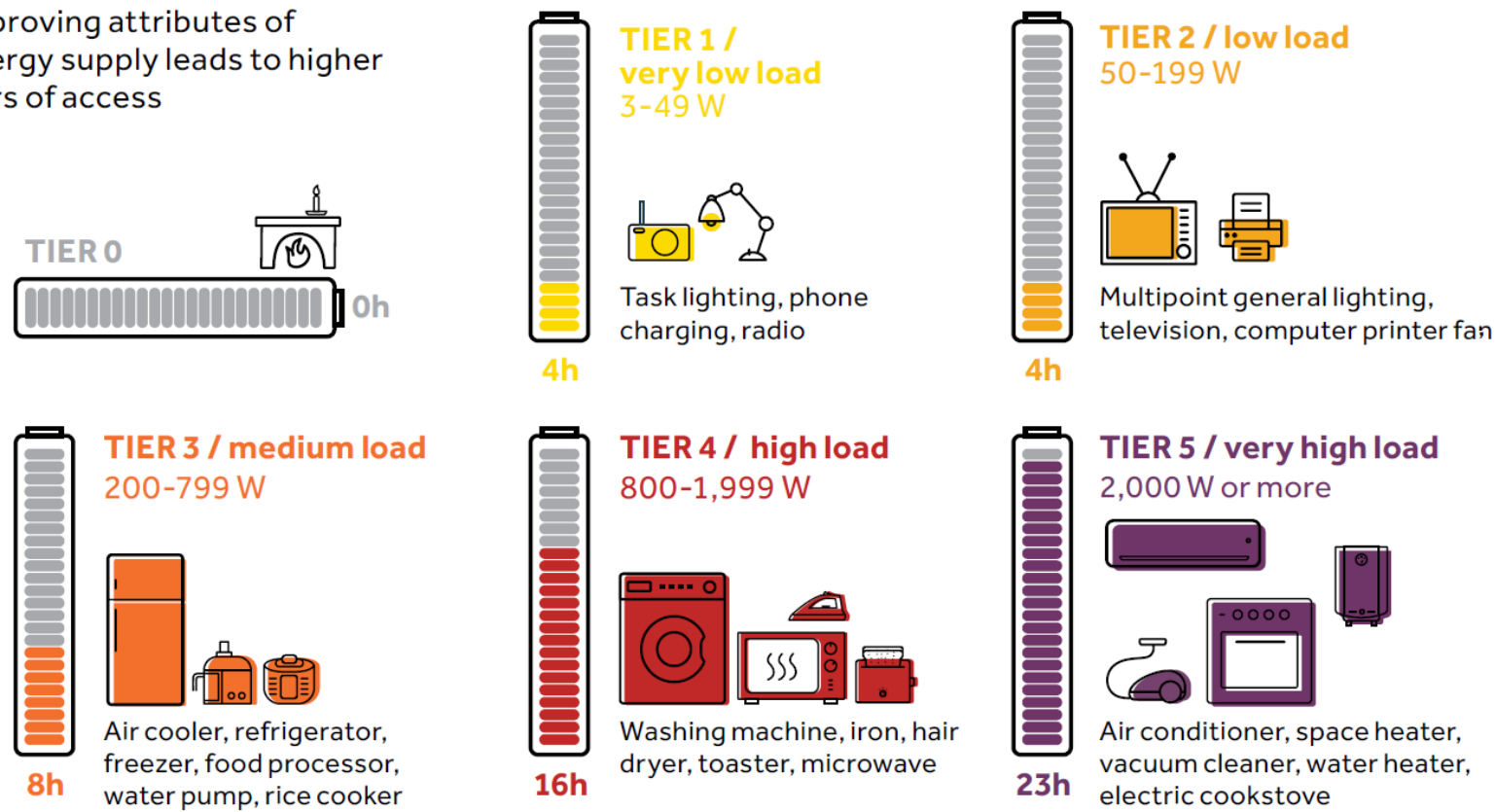
 High
  Low

Stopgap solutions deliver **poor lighting**, and are often **dangerous** and **expensive**

“Access” is no longer binary (whether a consumer is connected to an electricity grid): now based on “tiered” levels of service

Multi-Tier Framework (MTF) definition: “the ability to avail energy that is adequate, available when needed, reliable, of good quality, convenient, affordable, legal, healthy and safe for all required energy services”

Improving attributes of energy supply leads to higher tiers of access



Source: Adapted from the MTF, World Bank 2019.

Tiered levels of service delivered via varying technologies, with many stakeholders considering Tier 1 as a minimum threshold

	Tier Level	Electricity Source	GHG Impact	Upfront Cost*	O&M Cost
Quality-verified solar lantern 	1 for <hh	Solar PV			
Small solar home system (SHS) 	1	Solar PV			
Medium SHS	2	Solar PV			
Large SHS	3	Solar PV			
Mini-Grid 	2-4	Varies			
Grid 	5	Varies			

 High  Low

Standalone, mini-grid & grid provide connectivity

Scalable solutions, grow with household **needs** and **ability** to pay

Not all solutions are created equal when it comes to **climate impact**

More recent efforts have advocated for a more ambitious level of service, known as the “Modern Energy Minimum”



- **No high-income country** is low energy. Data shows a correlation between income and energy consumption at country level
- High-income countries have annual electricity consumption above **3,000 kWh** per capita
- One common threshold for “**energy access**” is modest, roughly equivalent to 50 kWh per capita in rural areas, and correlated with incomes of \$0.27 per day

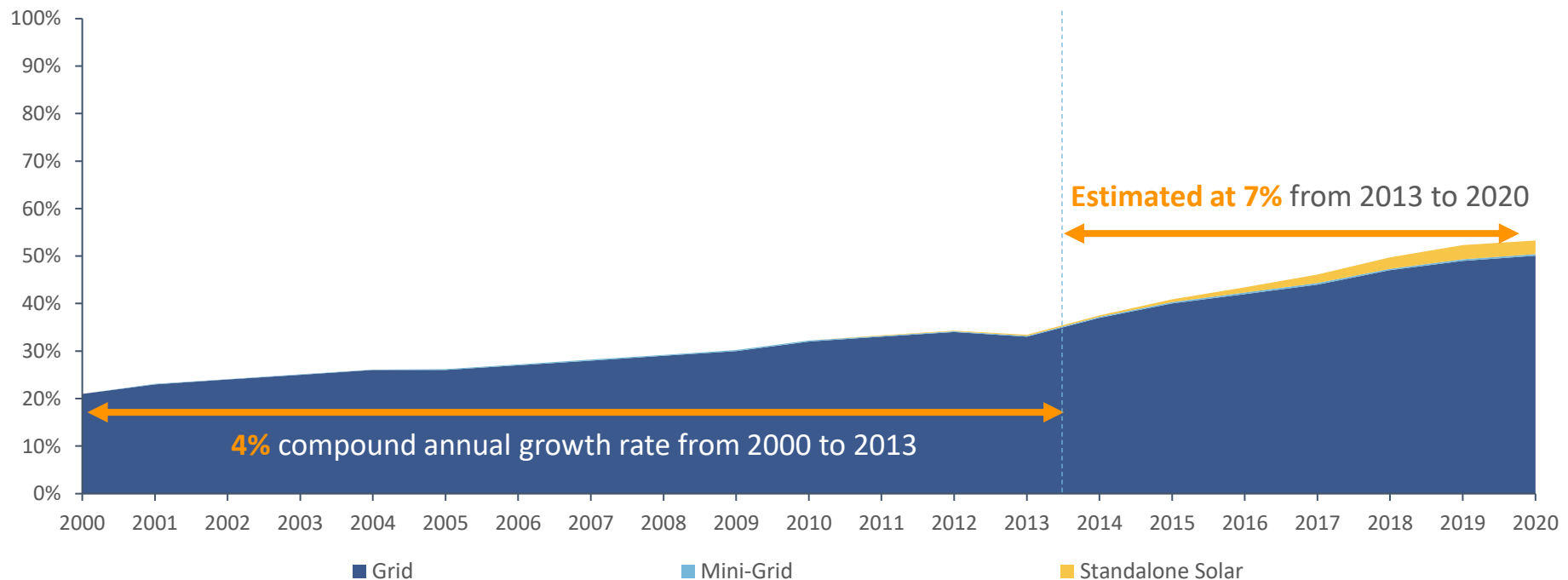
The **Modern Energy Minimum (MEM)** sets a target of **1,000 kWh** per capita per year

- Divided between 300kWh of household + 700kWh of non-HH electricity consumption
- Correlates with an income of \$6.85 per day

Meanwhile, Africa's electrification trends show some promising progress toward SDG 7

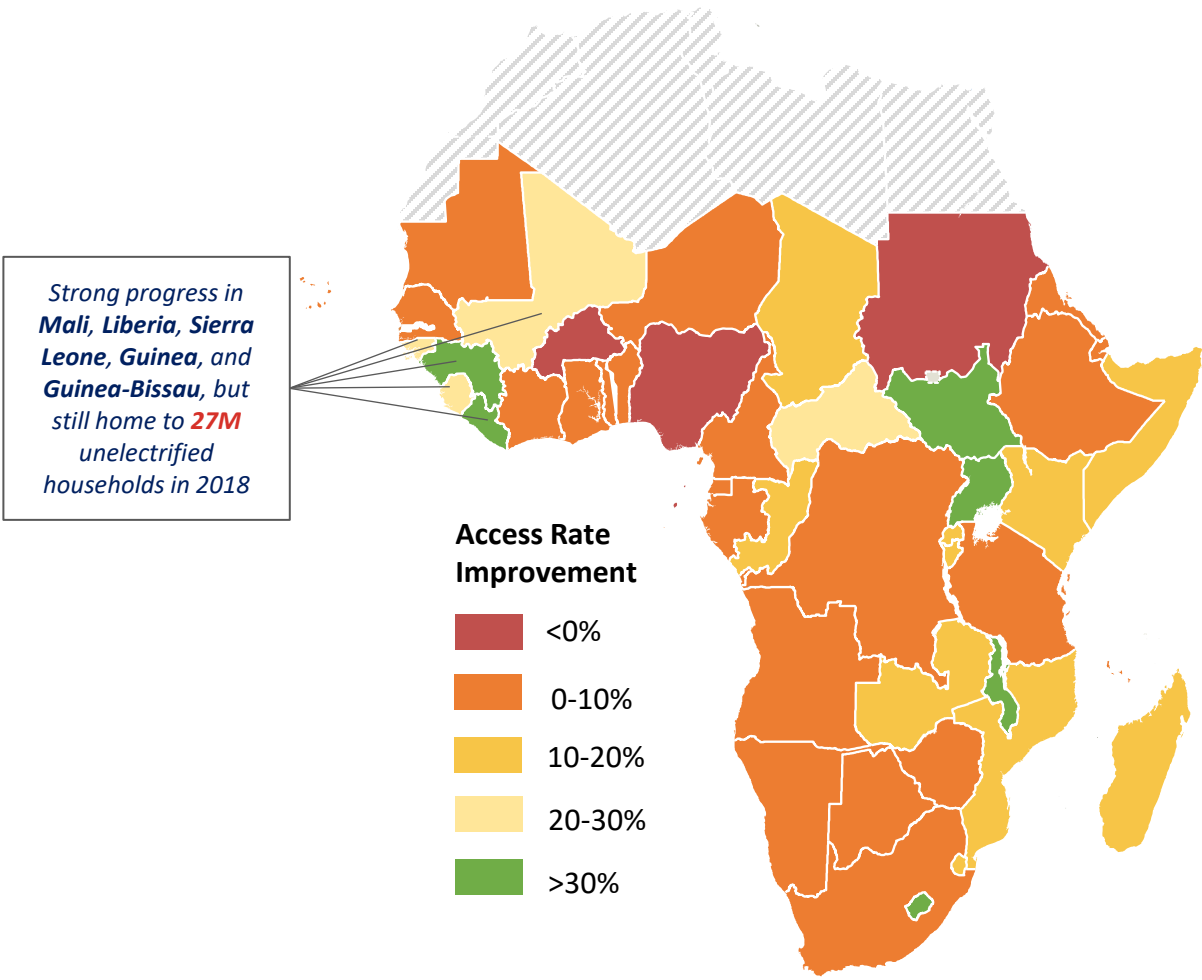
- Recent grid connection improvements tied to **grid densification**
- Significant contribution from standalone solar, particularly in East Africa
- Despite these trends, business-as-usual yields a significant SDG 7 shortfall

African Household Electricity Access by Type



Though this progress is very uneven across countries

Sub-Saharan Africa Household Electricity Access (2018 vs. 2016)



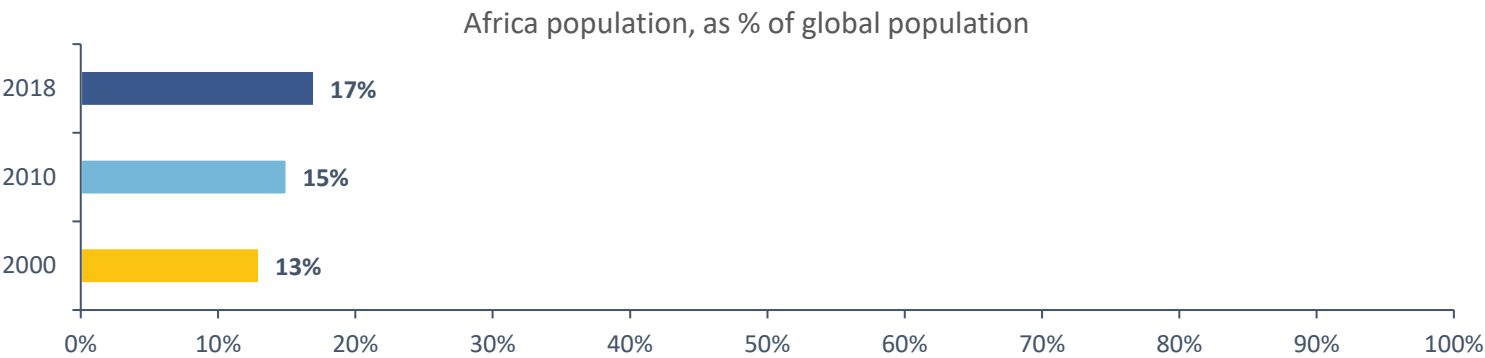
46 percent
median access rate for Sub-Saharan African countries in 2016

8 countries
have population growth that is outpacing the rate of new connections

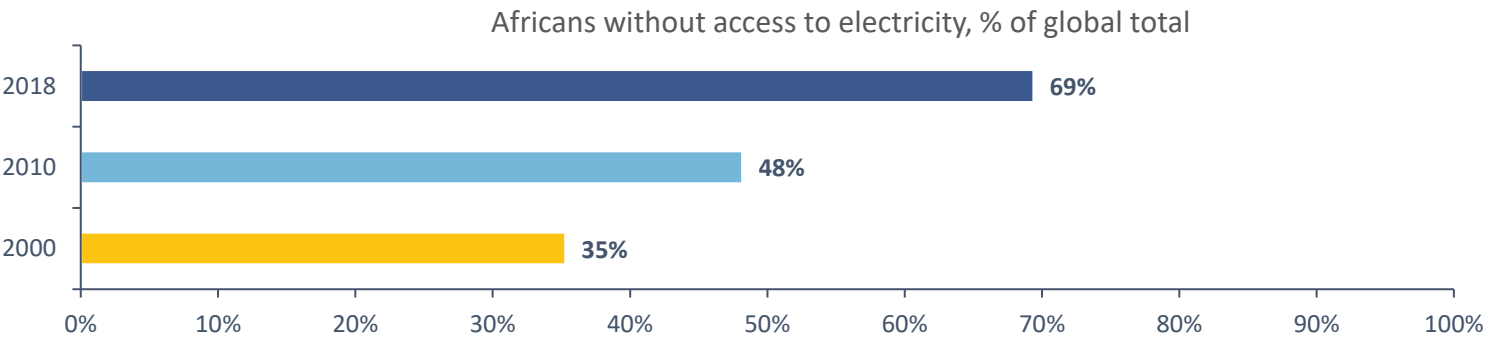
8 countries
have access in excess of **75%**, of which only 4 are not **small, island nations**

And Africa is falling behind rest of world on electricity access

Africa’s share of the global population has seen modest growth since 2000

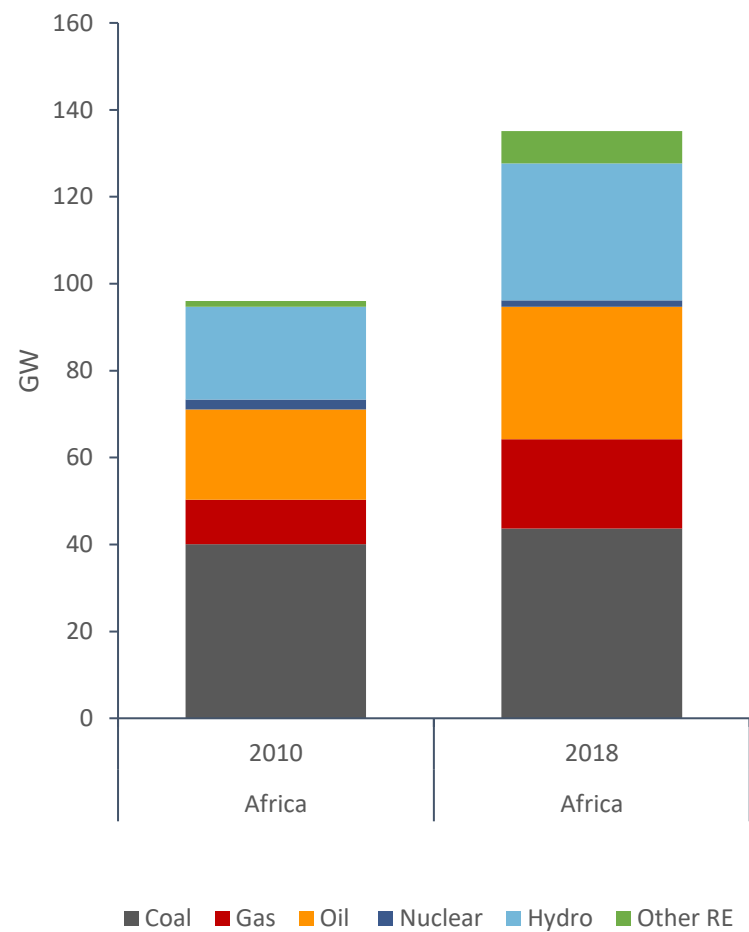


Yet it now hosts 69% of the world’s unelectrified households

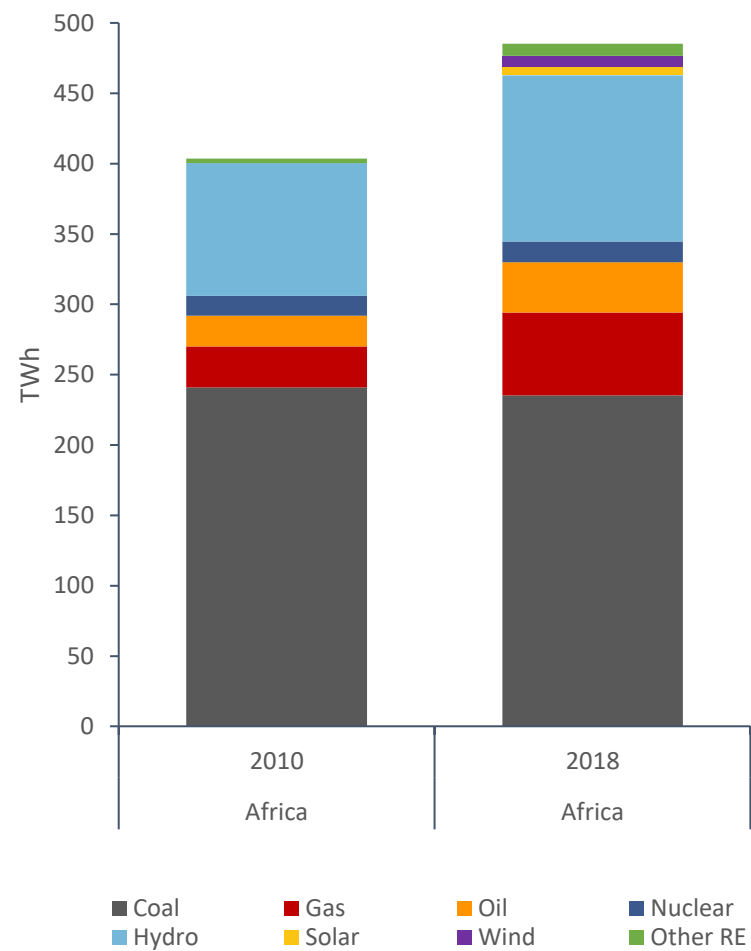


Africa's power grids remain heavily dependent on fossil fuel-based power

Installed capacity by fuel in Africa, 2010 and 2018



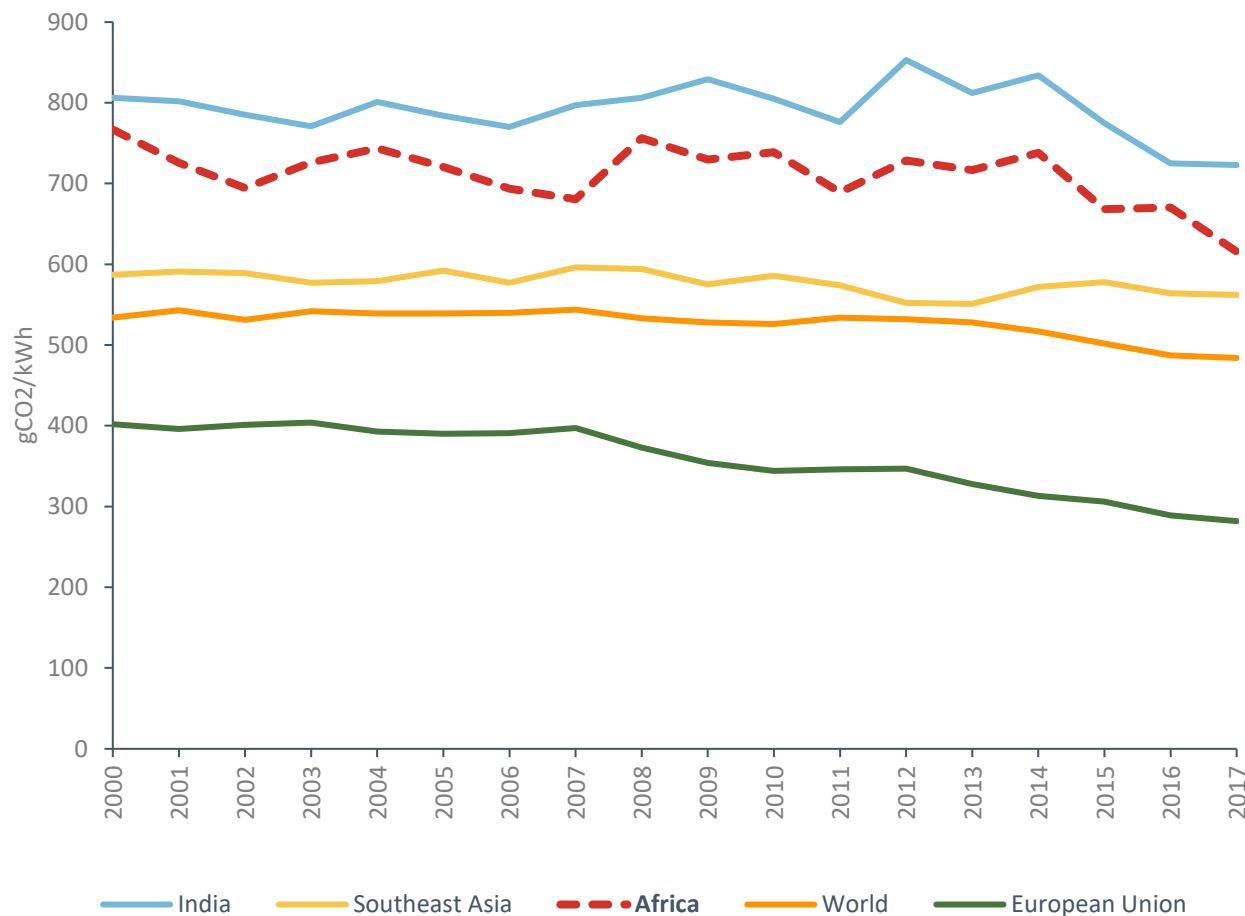
Electricity generation by fuel in Africa, 2010 and 2018



Source: Authors' estimations based on the IEA's 2019 World Energy Outlook and Africa Energy Outlook

And Africa's historical grid GHG emission trends have mirrored other parts of the world

GHG Intensity of Power Generation in Selected Regions

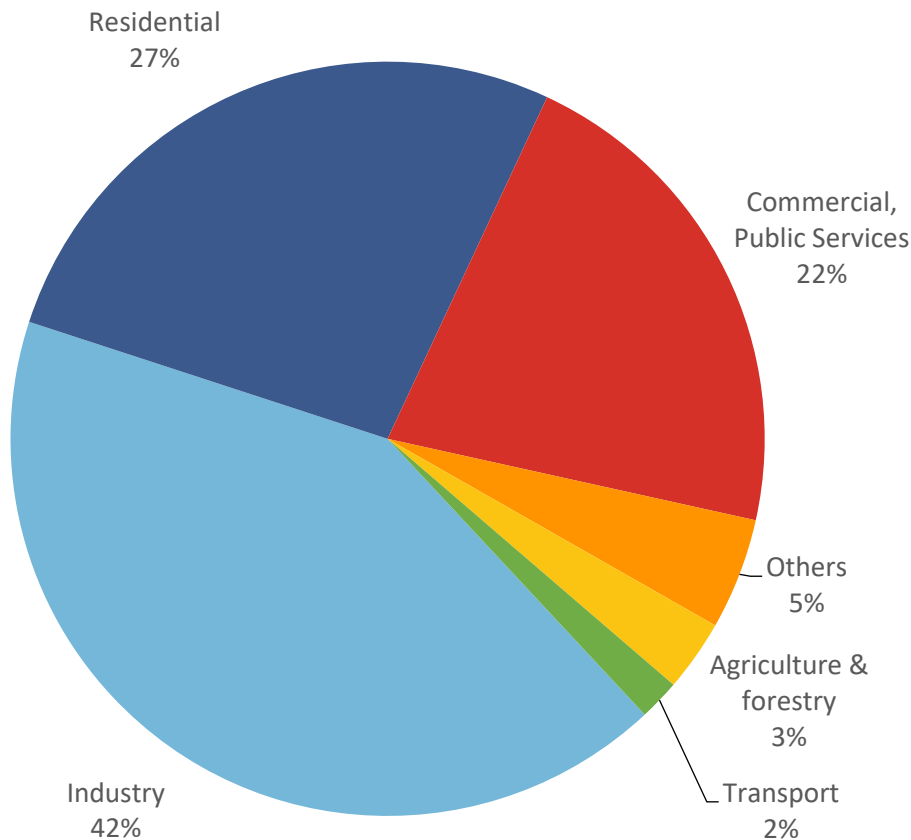


- Progress has been **inconsistent** in Africa, as seen by lumpy lines
- Africa emissions** intensity remains relatively high compared to the **rest of world**
- Future trajectory depends heavily on Africa's **generation mix** and the extent to which it draws upon indigenous renewable resources (e.g. hydro, solar, wind)

Africa's backup genset epidemic: high-cost power with a big climate impact

More so than households, enterprises are the major driver of electricity demand globally and in Africa

Global electricity consumption by sector, 2017



69 percent

of global **electricity demand** driven by **industrial and commercial** off-takers

While still nascent, **Africa's** commercial and industrial sectors similarly account for over **70% of total electricity demand**



27 percent

of global **electricity demand** originates from **households**



Despite lower access levels, households in Africa still account for **over 29% of the continent's electricity consumption**

Unreliable grid connections lead to massive use of fossil fuel-powered gensets, particularly by enterprises

As utilities struggle to keep up with growing electricity demand, grid reliability will likely worsen, exacerbating dependency on expensive, polluting backup generators

Unreliable grid connections

- In **developing countries**, unreliable grids are the primary driver for genset use:
 - ~**75 percent** of sites using fossil-fuel powered gensets are “**grid connected**”
- In **Africa** alone:
 - **36 countries** are considered to have unreliable grids¹
 - Enterprises experience an average of **9 outages per month**, lasting a total of 81 hours (e.g., > 10% downtime)
 - Unreliable grid connections result in an average **8 percent loss in business revenues**



Use of backup generators

- **Backup fossil-fueled generators** are used by households and enterprises
- **Powered with fossil fuels**, typically diesel or gasoline
 - Significant source of air pollutants
- Off-grid enterprises often resort to using gensets for power, particularly for **productive use applications**
- Some off-grid households use them as well, though fuel costs make them unaffordable for most



These gensets deliver power, but at a huge economic and climate cost

~ **7** million

estimated number of backup genset sites in Sub-Saharan Africa today, equivalent to **120 coal-fired power stations**

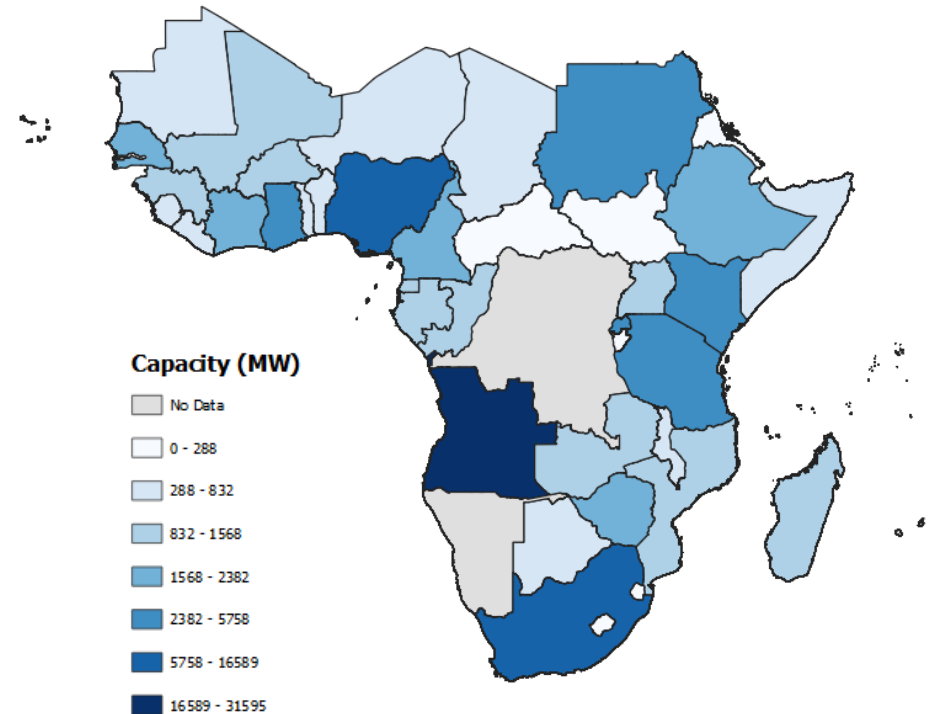
~ **US\$13** billion

spent in Sub-Saharan Africa by backup genset users each year on **fuel**

20 percent

of **gasoline & diesel** consumed in **Africa** is a result of backup genset use, equivalent to 15-20 percent of spending on education & healthcare

Installed capacity of back-up generators in Africa, 2016



Clean cooking: what it is and why it matters for climate change and energy access

Why clean cooking solutions matter

Four billion people - half of the world's population - depend on polluting solid fuels, open fires or inefficient stoves to cook their food¹

The annual costs are striking:

>4 million
premature deaths

1.4 billion
tons of **wood fuel** consumed, much of it
unsustainably

US\$38 – US\$40 billion
spent on **solid fuels** for cooking

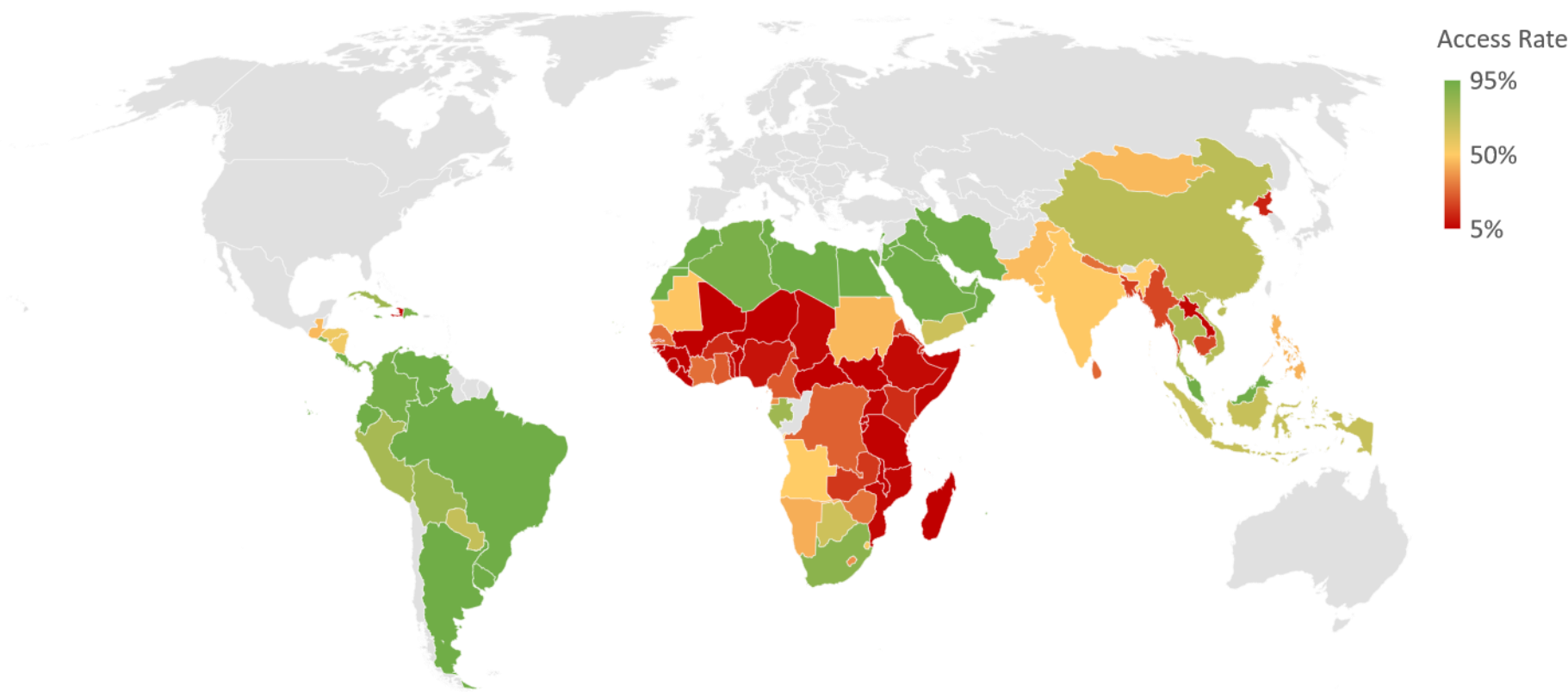
140 million
productive person-years **wasted on biomass**
collection



Africa is the epicenter of the cooking crisis

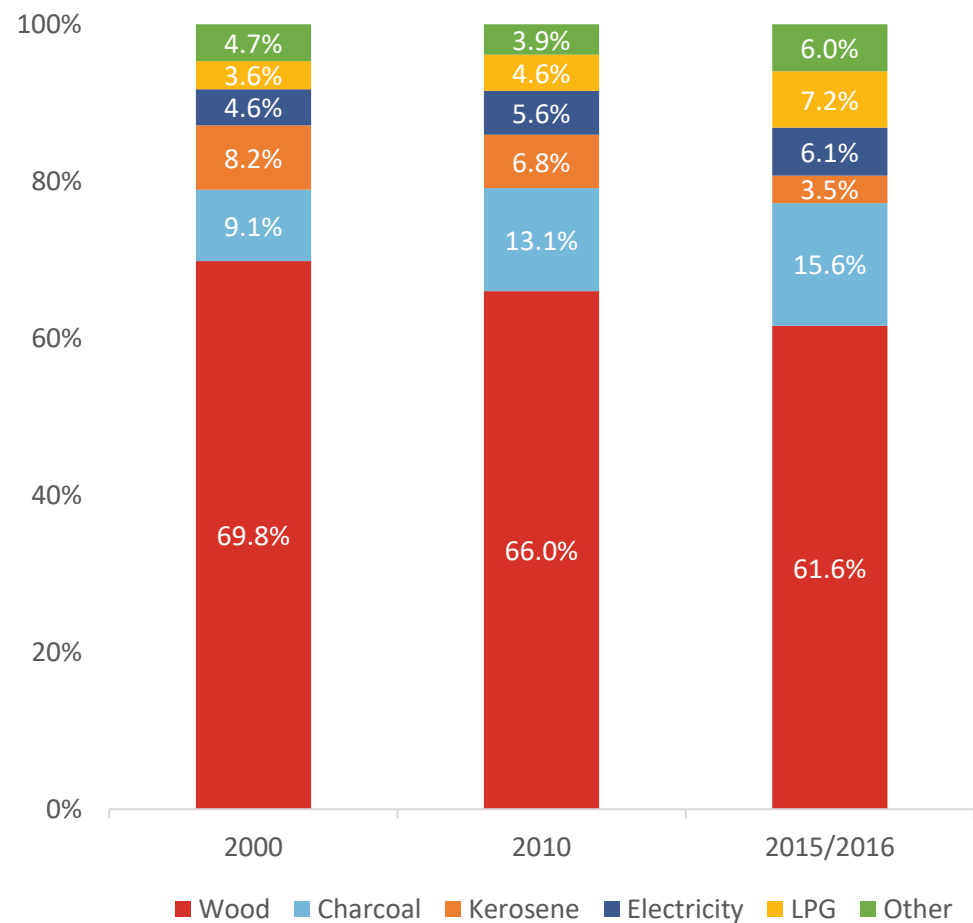
Population growth in Africa is **outpacing** annual growth in **clean cooking access** across the continent; between 2010-2018, the number of people without access rose from 750 million to 890 million

Percentage of population with access to clean cooking by country, 2018



Africa's high solid fuel use costs lives and the climate

Africa primary household cooking fuel use by type, 2000, 2010, and 2015/16 ¹



850 million+
Africans, or 80% of the continent's population, use **solid fuels** for primary cooking needs

600 thousand
Africans killed annually from household air **pollution**, making it the 2nd largest health risk in the region

600 Mt CO₂
of global **GHG emissions** comes from **solid fuel cooking** in Africa alone

Sources: Clean and Improved Cooking in Sub-Saharan Africa 2014, ACCES; The State of Access to Modern Energy Cooking Services, ESMAP 2020; Authors' estimates
¹ Other includes a mix of both modern and traditional fuels, including natural gas, biogas, mined coal, dung, crop waste, etc.

Solving the cooking challenge requires a shift to modern fuels

Making the shift must overcome significant barriers, including willingness and ability to pay, building fuel supply chains, and behavior change relative to traditional cooking approaches

	Traditional fuels				Modern fuels			
	Stopgap cooking		Improved Cooking		Modern Fuel		Renewable Fuel	
Stove Type	"3-stone"		Improved artisanal		"Standalone" stove		"Standalone" stove	
	Unimproved artisanal		Industrial		"Grid-tied"		"Grid-tied"	
Fuels	Wood		Wood	Coal	LPG		Electric*	Pellets
	Charcoal		Charcoal	Kerosene	Natural gas		Biogas	Ethanol
	Coal						Solar	
Example	Traditional Metal Stove	Chitetezo Mbaula	Burn jikokoa	Envirofit G-3300	LPG Stove Télia n°2	LPG/NG 2B SS gas stove	Mimi moto	SAFI Cooker
Image								
Fuel Type	Charcoal	Wood	Charcoal	Wood	LPG	Natural gas/LPG	Pellets	Ethanol
Efficiency	23%	20%	44%	34%	49%	N/A	47%	64%
GHG Emissions	Very high	High	Medium	Medium	Low	Low	Very low	Very low

Africa's choice of modern cooking transition depends on the emissions intensity of fuel sources

- In **Africa**, electric stoves have higher emissions than other fuel/stove combinations due to the **fossil fuel-heavy grid** generation mix
- The most GHG emissions-friendly solutions involve **lower-carbon fuels paired with highly-efficient stoves**

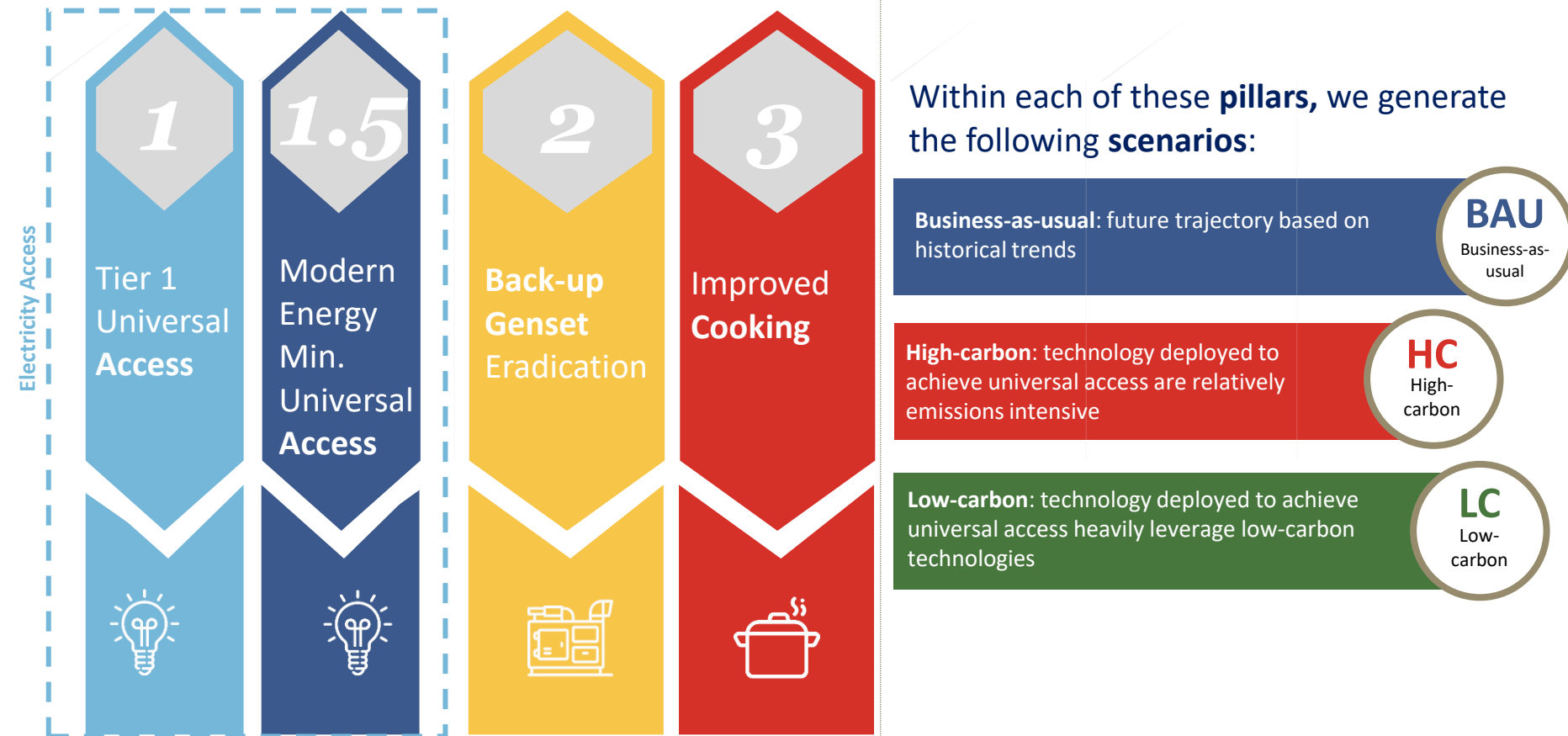
Fuel Type	Fuel Type	Fuel emissions intensity <i>kgCO₂e/GJ</i>	Stove Efficiency %	Stove and fuel emissions intensity <i>kgCO₂e/GJ</i>
Traditional stove	Charcoal	100	23%	434
Traditional stove	Wood	67	20%	337
Basic ICS	Charcoal	103	30%	343
Basic ICS	Wood	61	25%	249
Industrial ICS	Charcoal	103	41%	253
Industrial ICS	Wood	61	34%	181
Std. Kerosene	Kerosene	89	45%	198
Std. LPG	LPG	75	49%	153
Industrial Ethanol	Ethanol	24	64%	38
Industrial Pellet	Pellets	19	47%	41
Std. Electric	Electric	169	75%	226
Induction Electric	Electric	169	85%	199

**Each fuel has a GHG emissions intensity factor (kgCO₂e per gigajoule of fuel burned) which illustrates the carbon-intensity of the fuel when burned. When used in a particular stove, only a percentage of the burned fuel is converted into useful energy, resulting in a higher GHG emissions intensity in practice.*

Africa's energy access pillars and climate finance opportunities

Previewing the thematic pillars and climate scenarios

Our analysis is based on separate pillars, focused on:



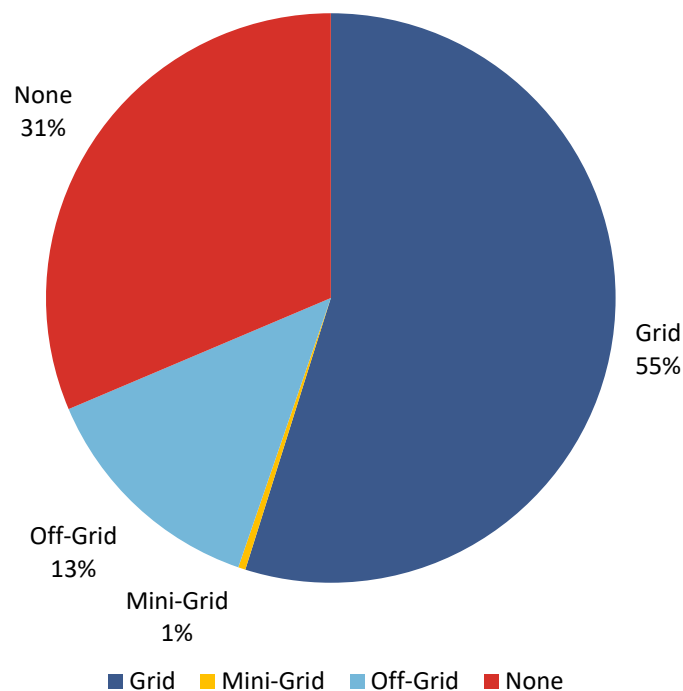
Each model allows benchmarking of scenarios to estimate avoided CO₂ emissions

Pillar 1: universal household electricity access by 2030

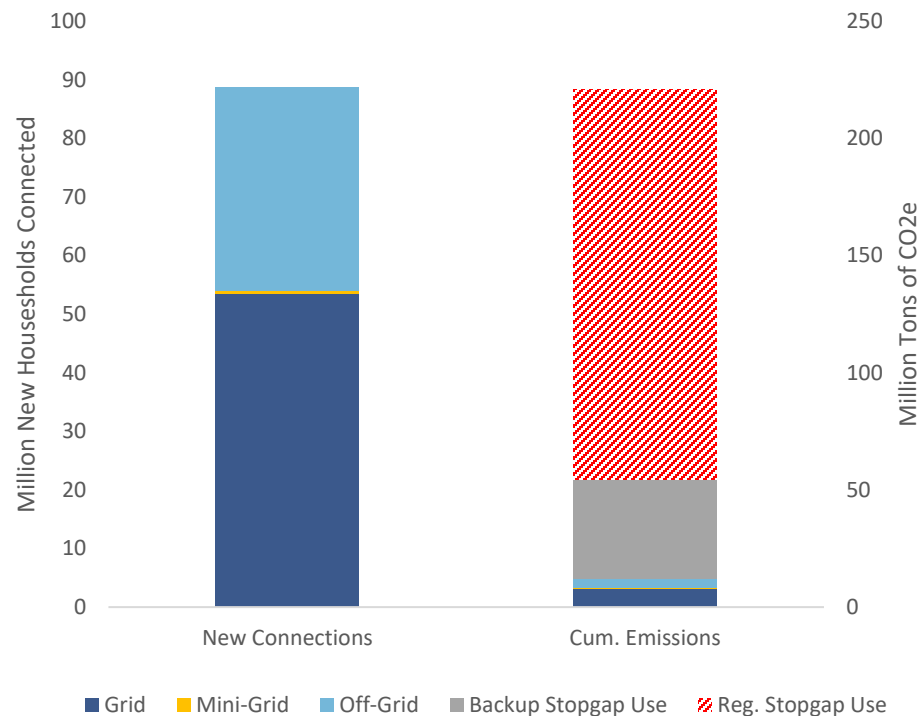
Pillar 1– Tier 1 Threshold

Business-As-Usual (BAU): model shows only 69% of HHs would have access in 2030¹; significant stopgap emissions footprint

Total HH Access in 2030



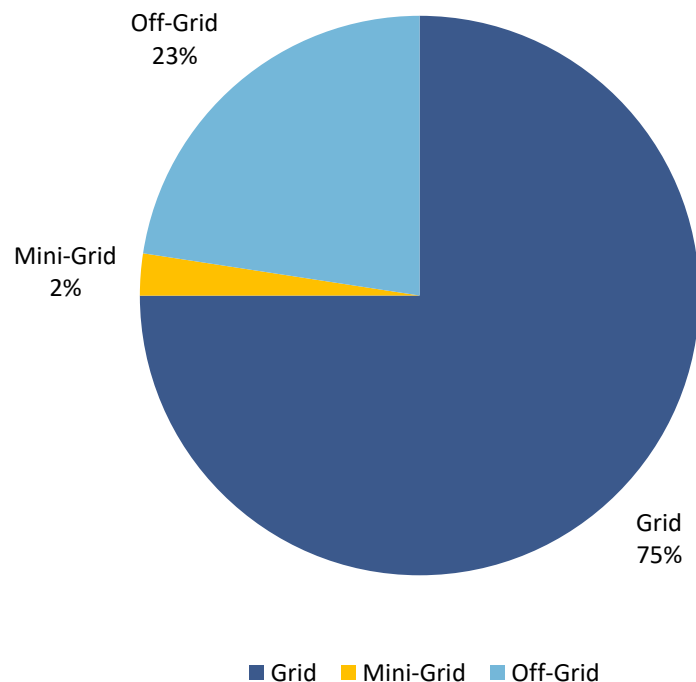
2021-2030 New Connections vs. Cumulative Emissions



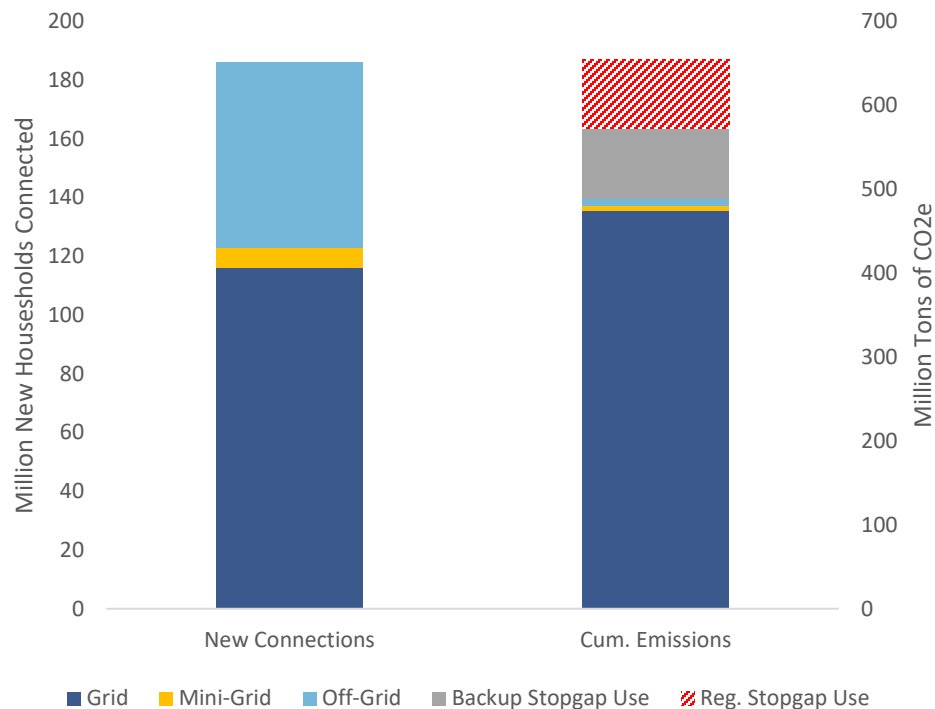
- Reflects expected access situation through 2030 if current trends in grid, mini-grid, and standalone solar access continue
- About **89 million households** will gain access; a **somewhat larger number** will remain **unserved**
- **Cumulative emissions of 418 MtCO₂e**, of which **50 percent from continued stopgap use**

Tier 1 high-carbon scenario: universal access, with a 53% increase in CO₂e emissions compared to BAU

Total HH Access in 2030



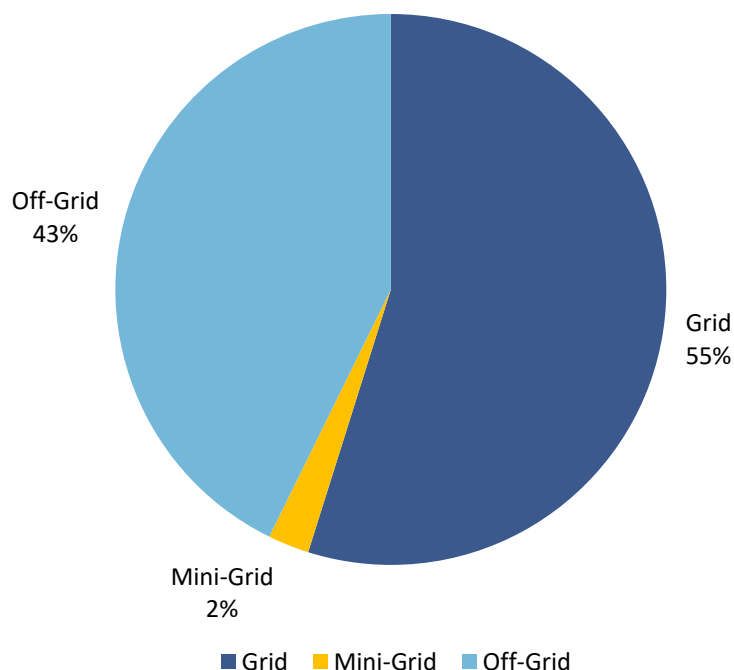
2021-2030 New Connections vs. Cumulative Emissions



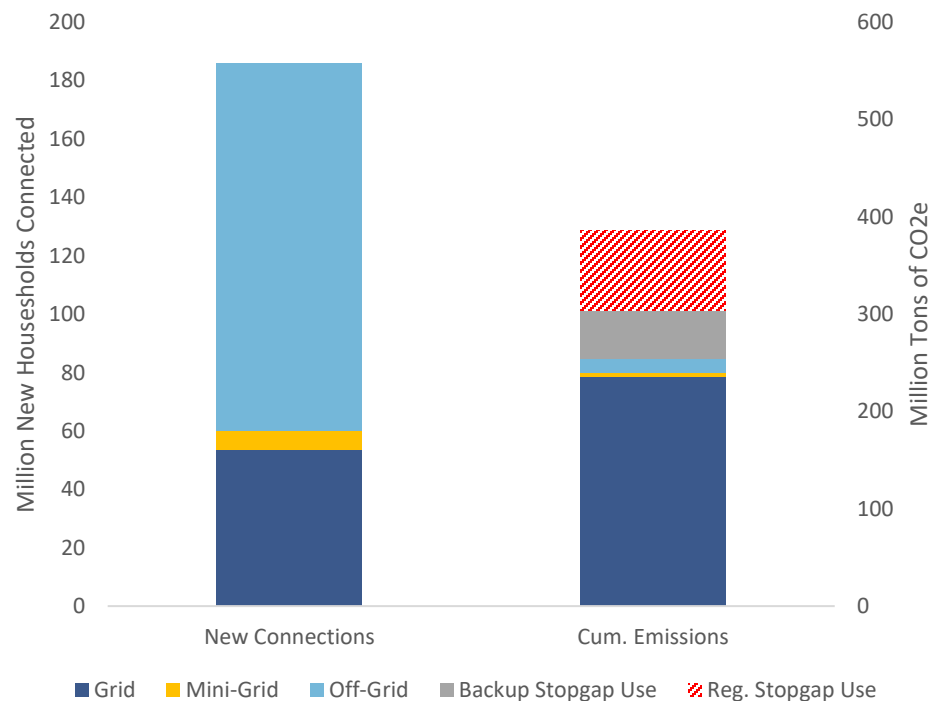
- **Universal access achieved** by 2030 through a **higher-carbon scenario** whereby the **grid** plays a large role, and associated emissions intensity rises to levels similar to **India today**
- Mini-grids and standalone solar also figure prominently, but the former continue to use fair amounts of **diesel**
- **Cumulative emissions of 654 MtCO₂e**, including an over 40 MtCO₂e reduction in stopgap emissions vs BAU

Tier 1 low-carbon scenario: universal access, smaller carbon footprint than BAU

Total HH Access in 2030



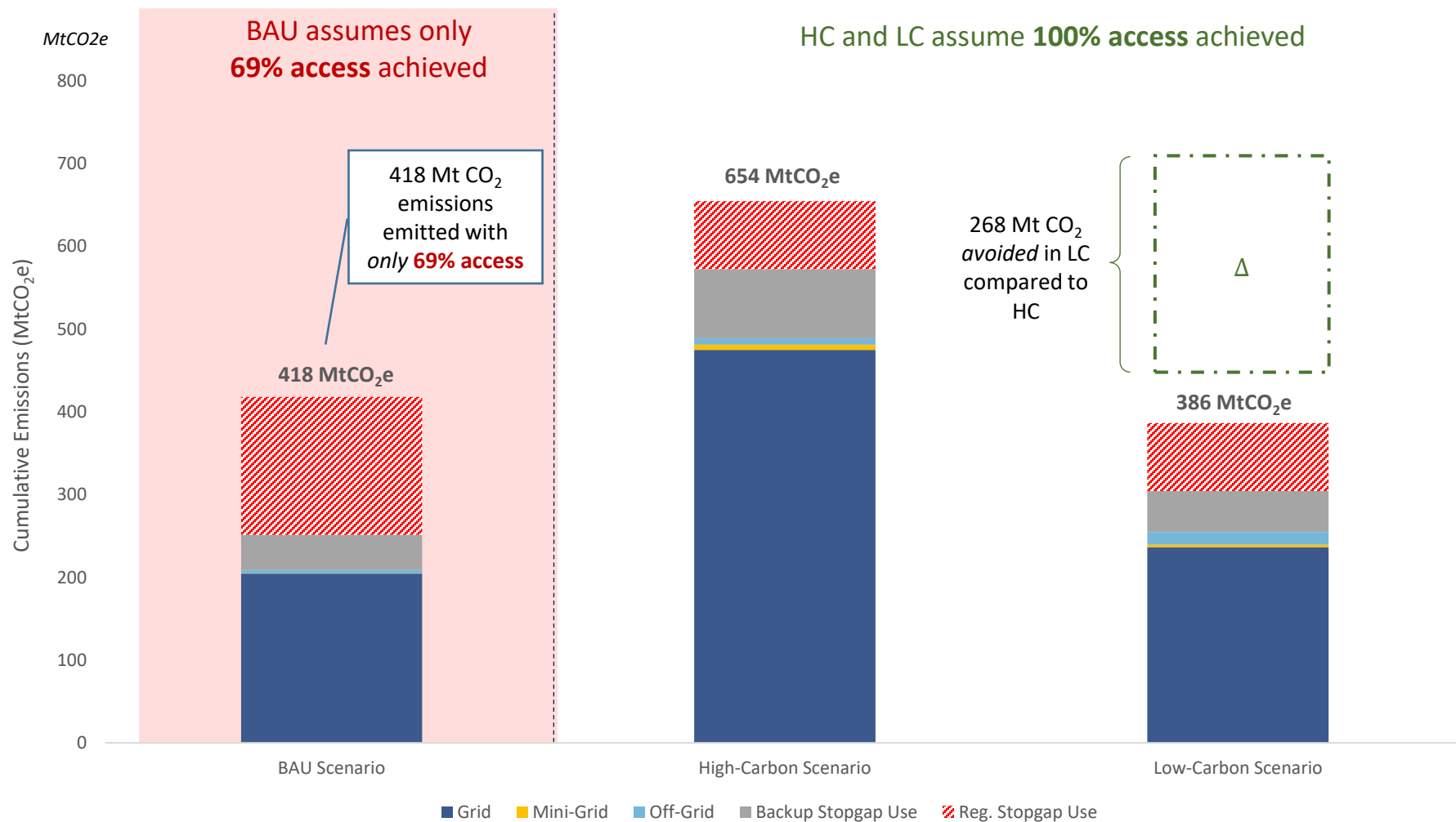
2021-2030 New Connections vs. Cumulative Emissions



- **Standalone solar, green mini-grid solutions** account for **majority** of new **household connections** through 2030
- Lower dependency on grid generation; **emissions intensity decreases** in line with historical trends;
- **Limited GHG footprint** of **off-grid technologies** means that emissions impacts relative to BAU are rather **limited**
- **Cumulative emissions of 386 MtCO₂e**, including an estimated 77 MtCO₂e avoided stopgap emissions vs BAU

Emissions impact of universal access scenarios vary widely: low-carbon scenario has net avoided emissions of 268 Mt CO₂e

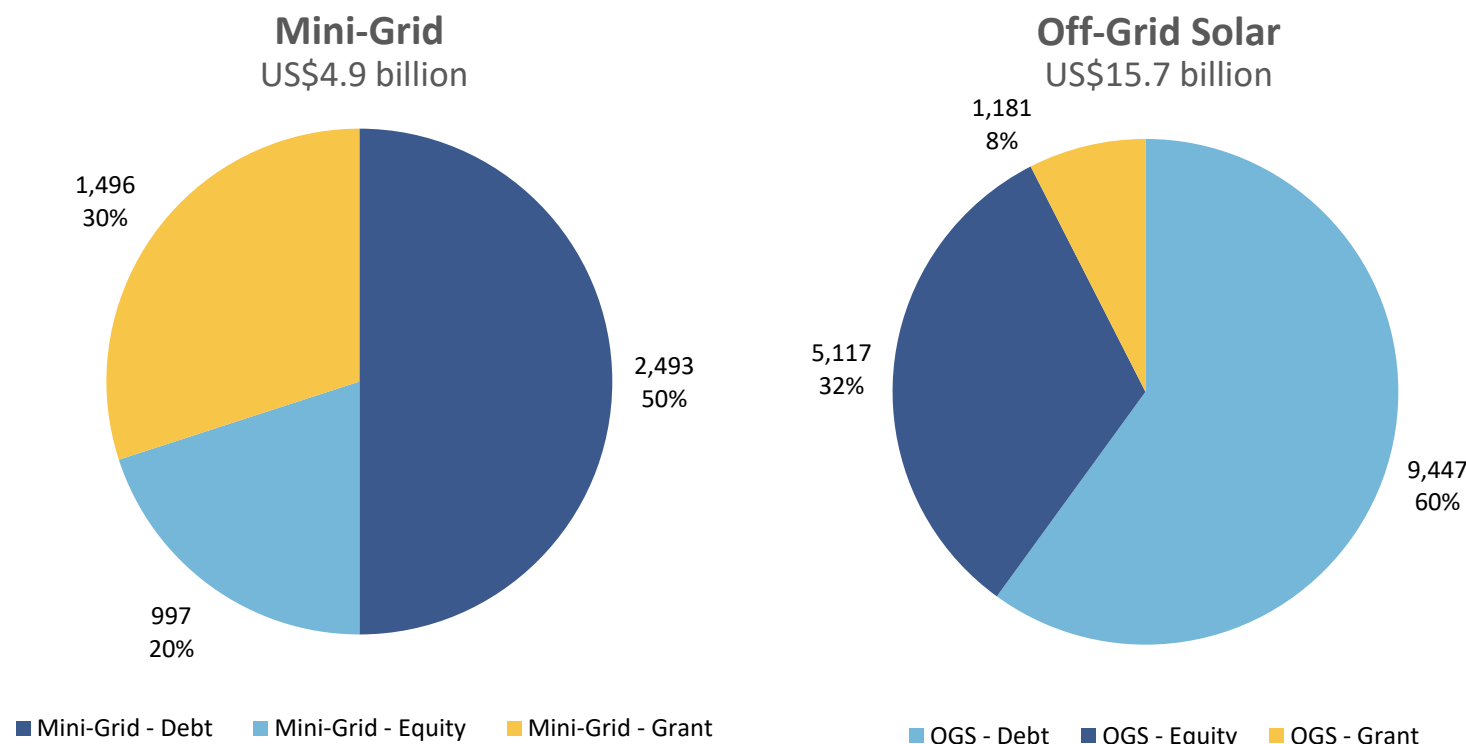
Comparison of 2021-2030 Cum. CO₂e Emissions



Tier 1 low-carbon electrification scenario yields a US\$20.7 billion climate finance opportunity

- **Off-grid solar** financing need of **US\$15.7 billion**, yielding **first time access** for **125.7 million** households
- **Mini-grid** financing of **US\$4.9 billion** required, delivering access to **6.8 million** households

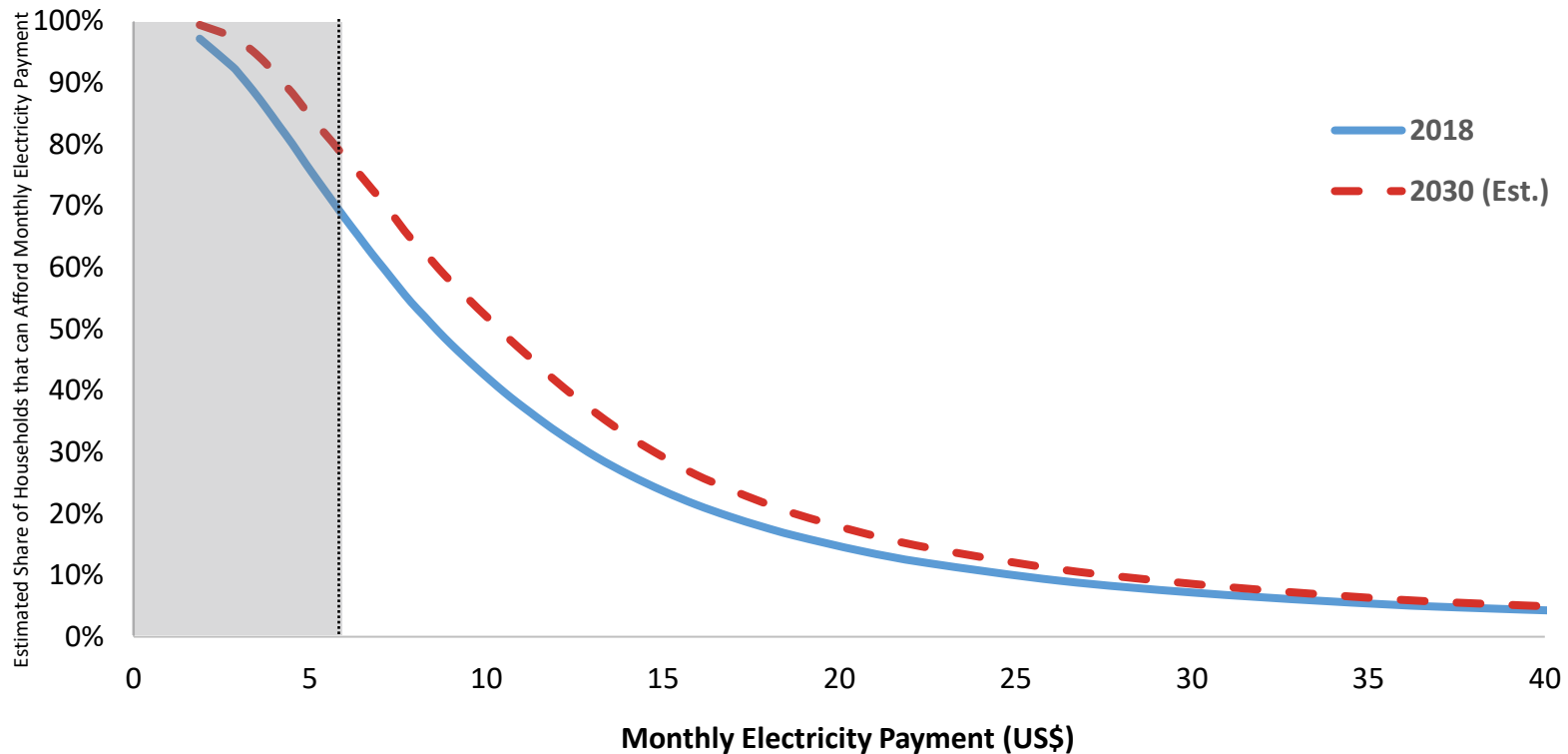
Base Demand: Off-Grid Solar & Mini-Grid Capital Needs, 2021-2030



US\$2.5 billion required in consumer subsidies to achieve universal access under low-carbon scenario

- **42 million households** that need but cannot afford T1 OGS by 2030
- **31% of households** without grid or micro-grid connections that require subsidy for T1
- **\$2.5 billion** total financing required if 50% demand-side subsidy is provided

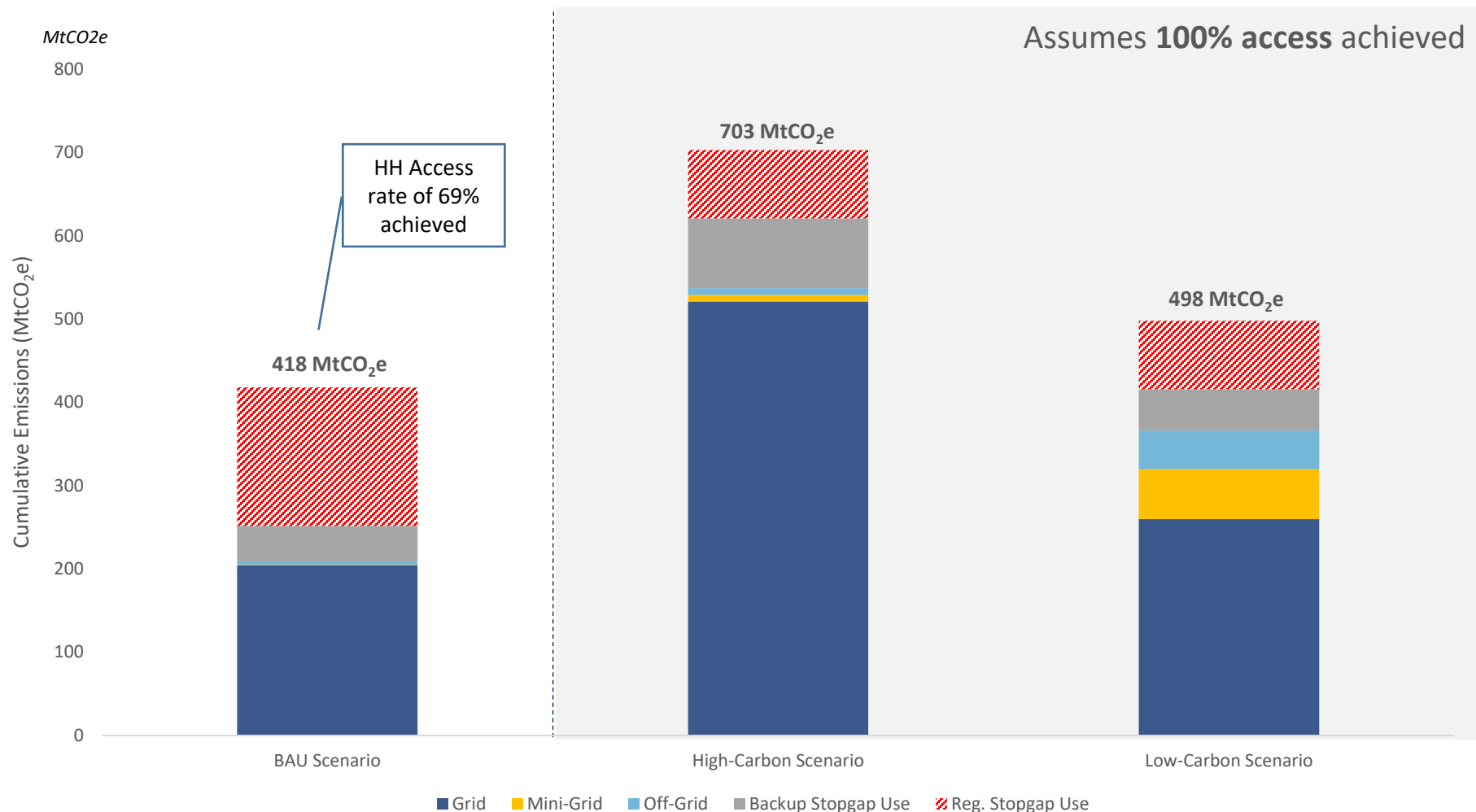
Sub-Saharan Africa Affordability of Monthly Electricity Payments



Pillar 1.5– Modern Energy Minimum Threshold

MEM yields additional 112 Mt CO₂e (compared to Tier 1), though per capita consumption increases to 130 kWh/year

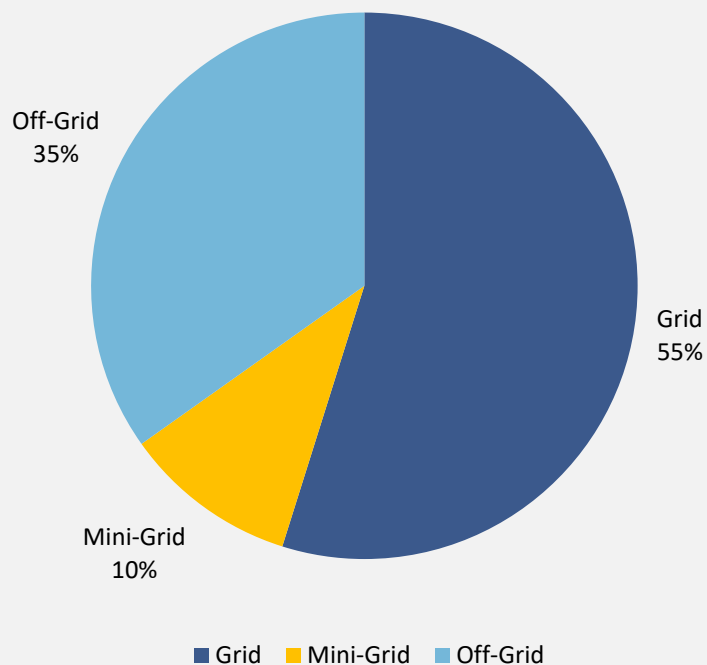
High Demand: Comparison of 2021-2030 Cum. CO₂ Emissions



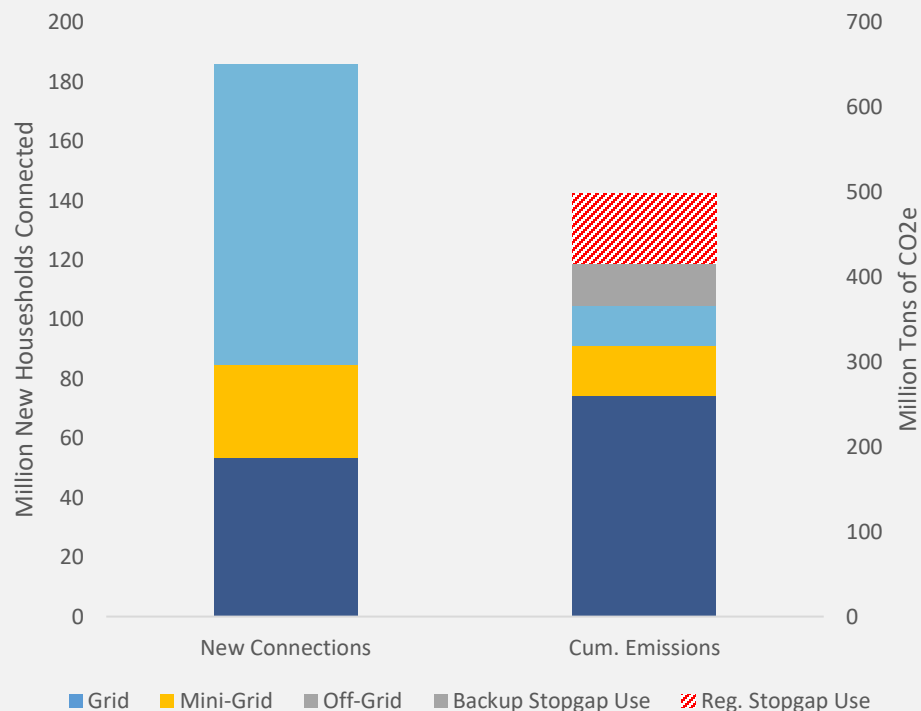
Source: Catalyst predictive model outputs

MEM requires 25M more mini-grid and 37M more Tier 3 OGS connections relative to Tier 1

Total HH Access in 2030



2021-2030 New Connections vs. Cumulative Emissions

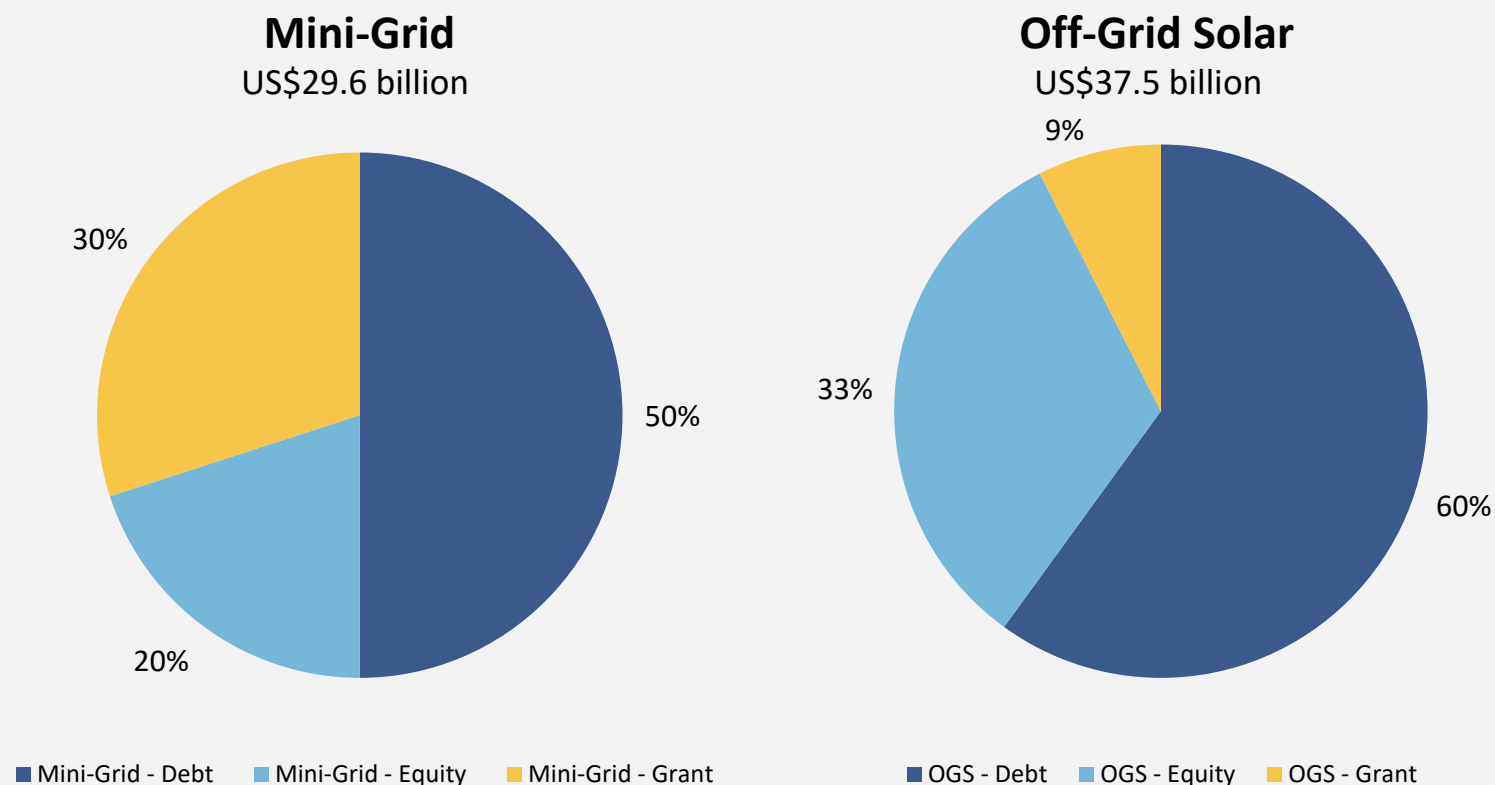


- MG and Tier 3 systems required to deliver on scenario's assumed per capita consumption of 130 kWh p/a by 2030
- Over the period, mini-grids account for 16 percent of new connections and Tier 3 OGS nearly 20 percent
- In the MEM scenario, average per capita electricity consumption increases by 57 percent, and emissions 27 percent, relative to the base case

MEM's low-carbon electrification scenario yields a US\$67.1 billion climate finance opportunity

- Larger systems required to deliver **higher per capita demand of 130 kWh** per year by 2030
- **Off-grid solar** connects **101 million** households at a cost of **US\$37.5 billion**, while mini-grids require **US\$29.6 billion** to electrify **31.3 million households**

High Demand: Off-Grid Solar & Mini-Grid Capital Needs, 2021-2030

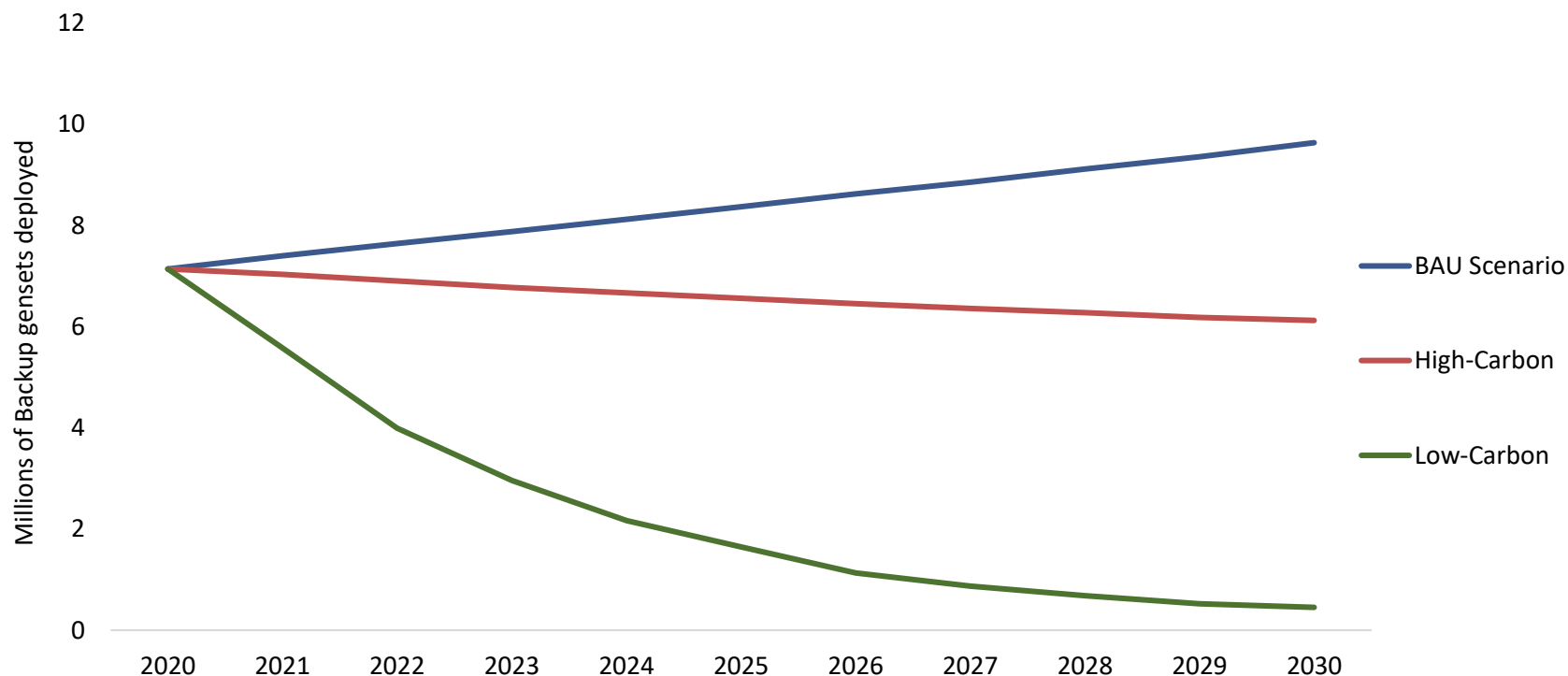


Pillar 2: eliminating backup genset use by African enterprises and households by 2030

The pace of phasing out fossil-fuel burning gensets with renewable solutions impacts scenario emission profiles

- **High-carbon scenario** assumes that, when gensets reach end of life, 10 percent are replaced with renewables
- In **low-carbon scenario**, at end of life, 50 percent of gensets are **replaced** with renewables; by 2030 nearly all gensets are retired

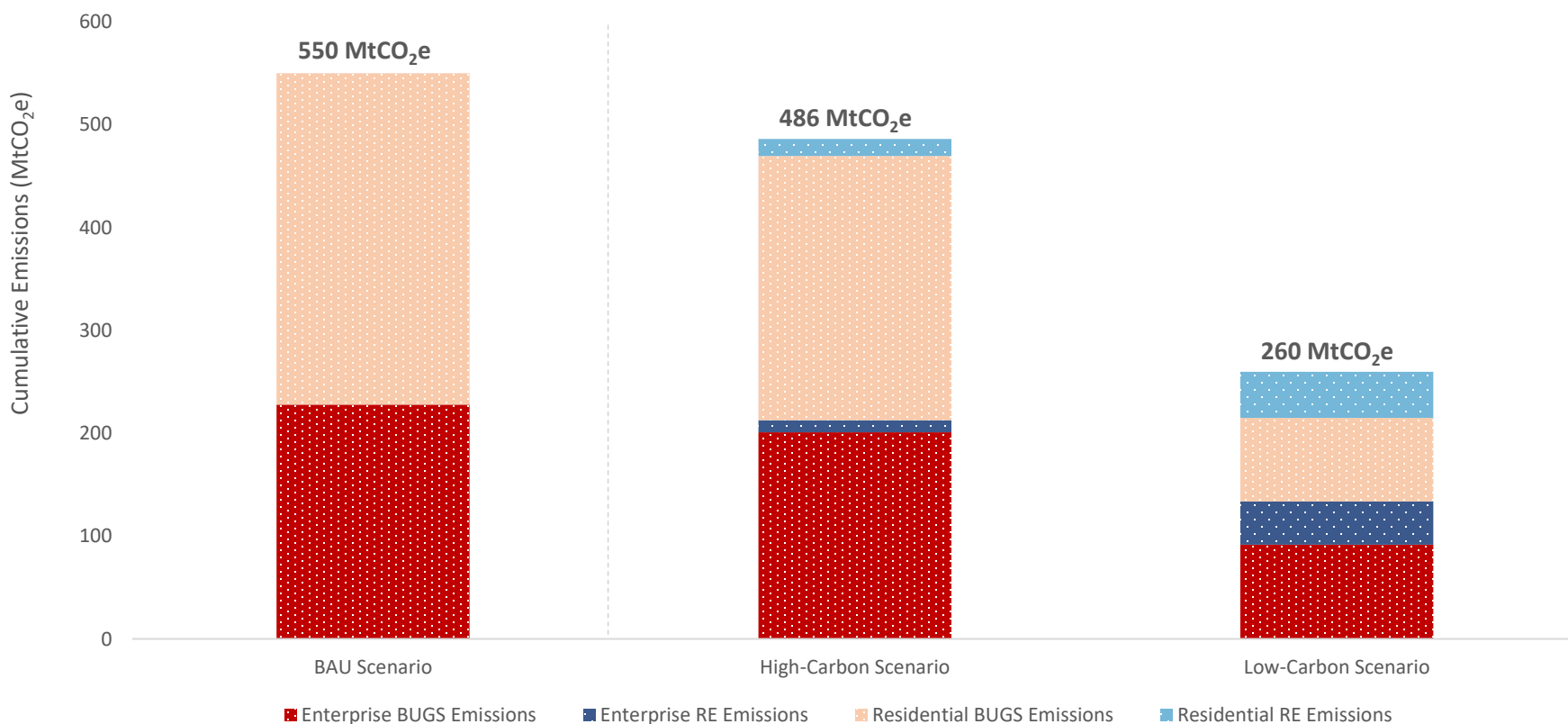
Evolution of Backup Genset Fleet for Households and Enterprises by Scenario, 2021-30



Displacing 9.2 million backup gensets with decentralized renewables reduces African emissions by 226M tons of CO₂e

- Emissions reductions are driven by the growth rate in backup genset fleets and the rate at which **backup gensets are replaced by** renewables; replacement rates are varied across scenarios
- BAU** has highest carbon footprint because renewables do not become part of the fleet mix

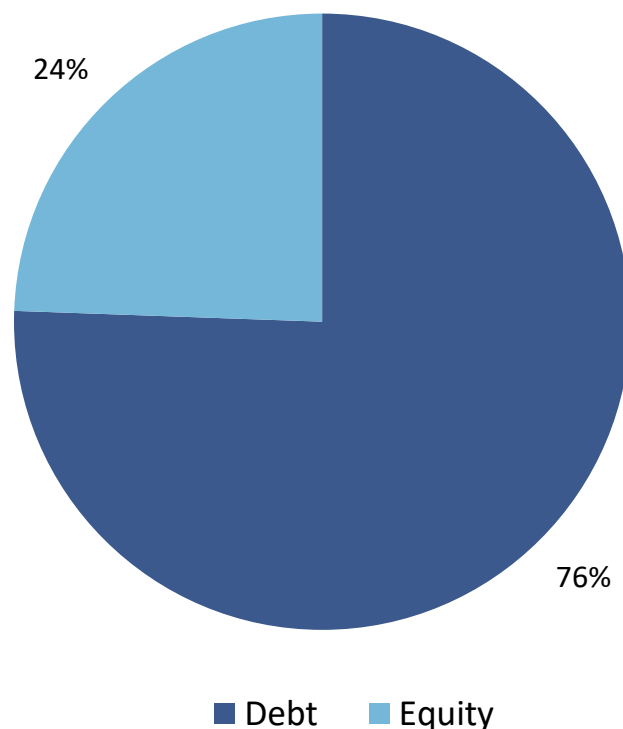
Comparison of 2021-2030 Backup genset, Household, and Enterprise CO₂ Emissions



Low-carbon scenario yields a US\$134.4 billion climate finance opportunity

- Displacing **9.2** million backup generators with a total generation capacity of **163 gigawatts** comes at a significant cost
- Generator displacement in later years is forecasted to be considerably cheaper thanks to **sharp reductions in RE technology costs**, particularly lithium-ion batteries

Back-up: Renewable Energy Capital Needs, 2021-2030



Pillar 3: moving African households that cook with charcoal onto clean cooking solutions

Clean Cooking: establishing a plausible 2030 scenario for Africa

What

- Achieving universal clean cooking in Africa by 2030 **not feasible**
- This research posits a plausible low-carbon cooking scenario whereby African **charcoal users transition to modern fuels**

Why

- Household's using **charcoal** are most likely to switch to **modern fuels** because:
 - Charcoal users have an **existing willingness and ability to pay** for fuel
 - Charcoal **has established distribution channels** that could be used for modern fuels
 - Charcoal users are typically in urban or peri-urban areas, thus **easier to access** and in greater density

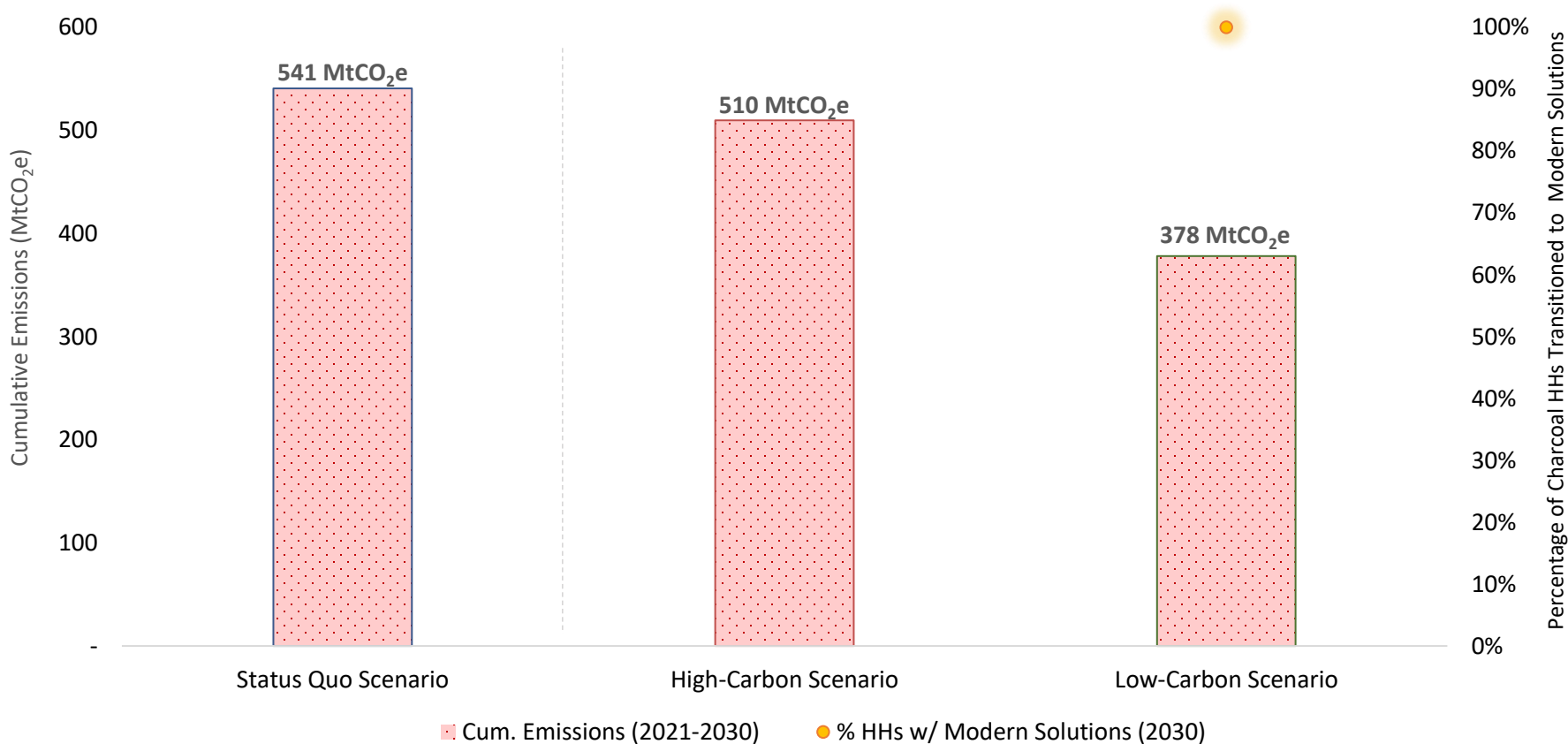
How

- Our cooking model estimates the impact of **moving charcoal users onto modern fuels**, including LPG, electricity, ethanol, and wood pellets
 - Scenarios vary by the percentage of charcoal-using HH that shift to modern fuels by 2030
 - High-carbon assumes 25%; low carbon assumes 100%; black carbon emissions are not included

Displacing charcoal with modern fuels in Africa yields 132 million tons of avoided CO₂e emissions

39 million new households would cook with **modern fuels**, representing an 83 percent increase in primary¹ modern fuel users in Africa

Comparison of 2021-2030 Cum. CO₂e Emissions by Scenario from Cooking



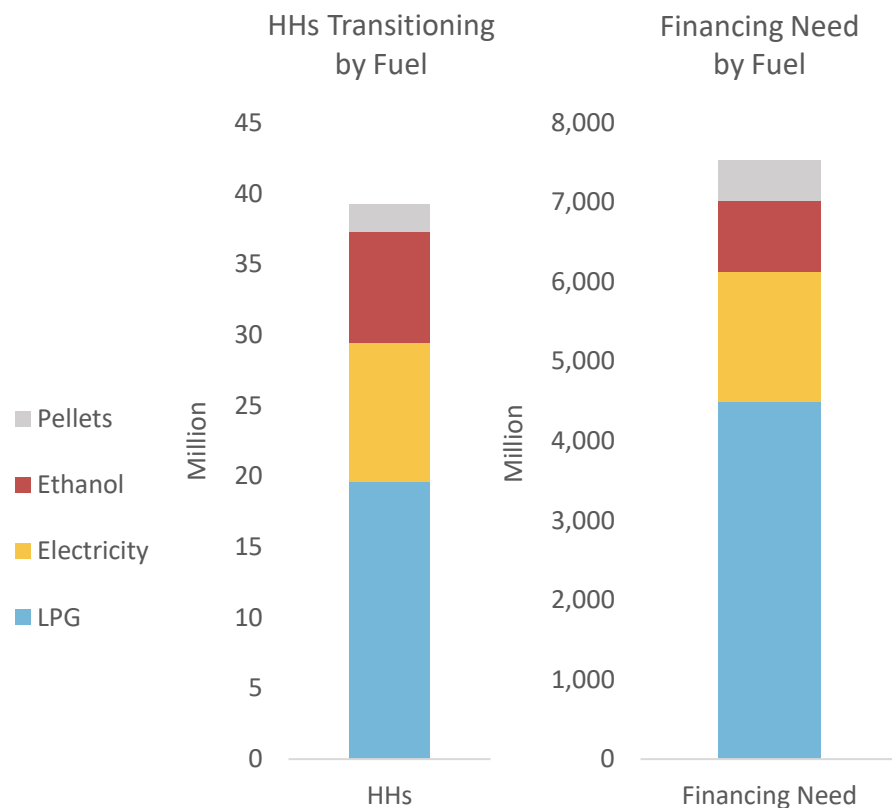
¹ In a household, the primary fuel is the one which accounts for the majority of cooking needs.

Displacing majority of charcoal with modern fuels in Africa would cost US\$7.5 billion

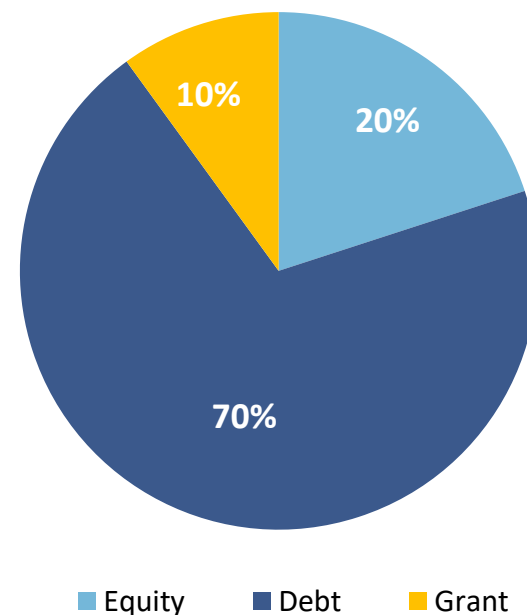
The majority of charcoal households transition to LPG and electricity
(*though stacking continues*)



\$7.5 billion investment required to produce and distribute stoves and build downstream infrastructure



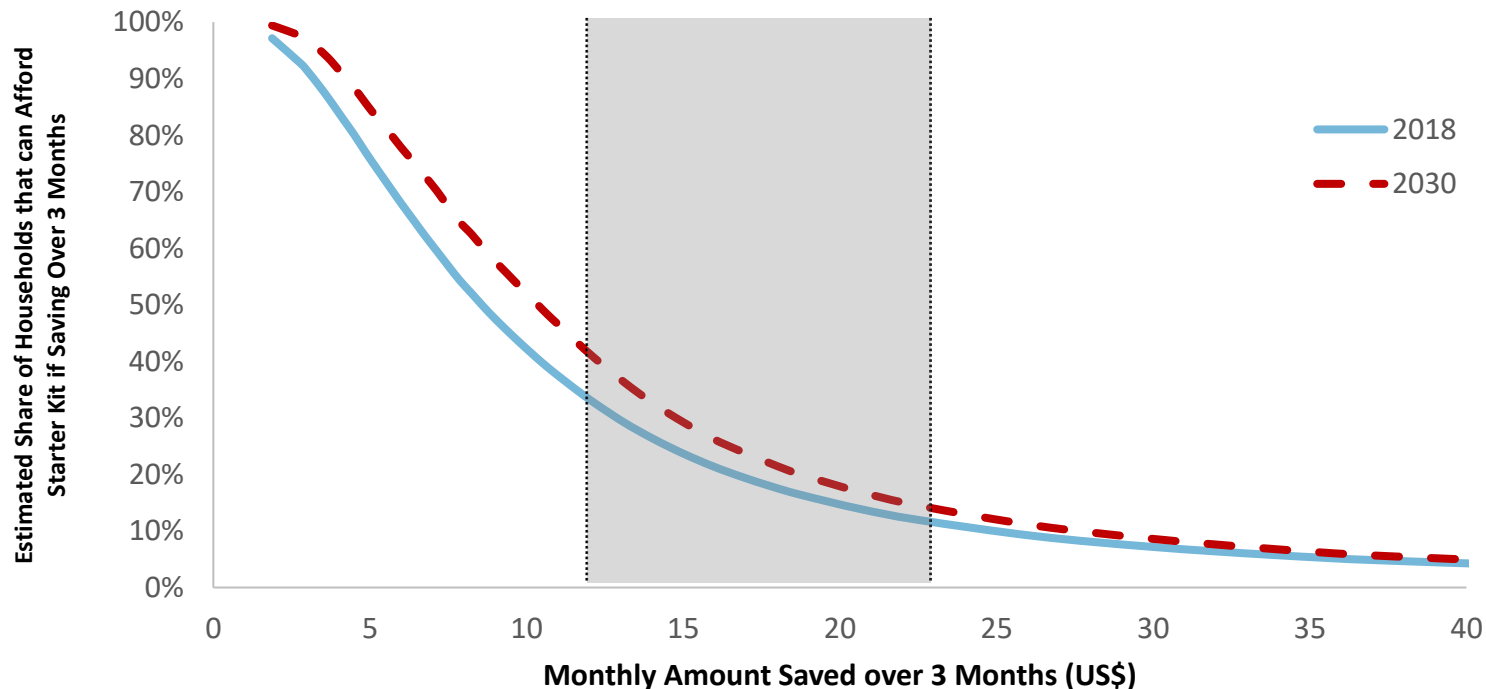
Clean Cooking Capital Needs, 2021-2030



The upfront cost of converting to modern fuels is high, ~\$70 for typical 'starter kits' (e.g., stove, cylinder, etc.)

- Moving the **39 million** households cooking primarily with charcoal today onto modern fuels will require a **subsidized starter kit** or **PAYGO modalities**.
- We estimate that without widespread PAYGO modalities for the various modern fuels, an upfront subsidy of 50% of the equipment cost, totaling \$2 billion over 10 years will be required to move charcoal users onto modern fuels.

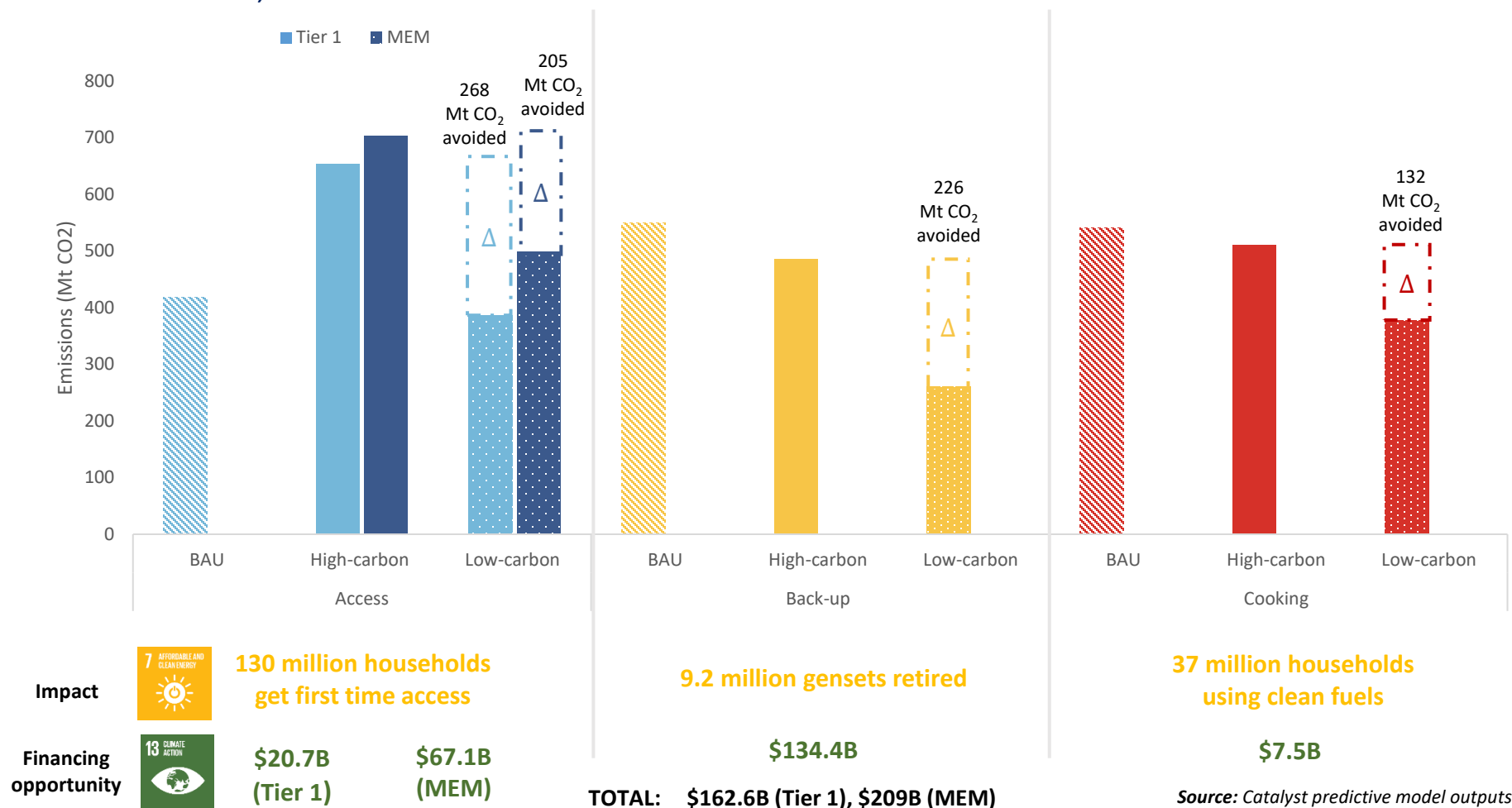
Share of households that can afford starter kit based on monthly savings over 3 months



Africa's SDG 7 and 13 call to action

These scenarios avoid 563-626 million tons of CO₂, deliver significant SDG 7 impacts, and unlock a substantial climate finance opportunity

Emissions from plausible low-carbon scenarios are benchmarked against equally plausible high-carbon counterfactuals; the difference between the two constitutes the avoided emissions



Summing up Africa's low-carbon access scenarios: huge impact, significant avoided CO₂ emissions, large climate opportunity



A **low-carbon scenario** contributes massively toward **universal access** and **improved cooking**



132 million

new connections from **off-grid** technologies delivered

9.2 million

gensets used by enterprises and households **displaced**

39 million

new households would **cook with modern fuels**



A **low-carbon scenario** benchmarked against a high-carbon scenario **yields**



563-626 million

tons of **avoided CO₂e emissions** over the next decade



A **low-carbon scenario** requires substantial volumes of **new capital**



US\$162-209 billion

climate finance opportunity

Three calls to action to climate-first investors to help catalyze the SDG7- climate finance nexus

1 Finance existing energy access enterprises

2 Support new, innovative mechanisms

3 Help define the next wave of investment opportunities



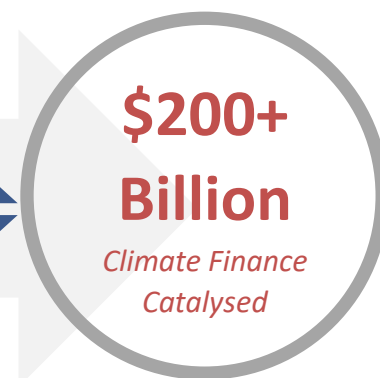
via existing vehicles (e.g. **CrossBoundary Energy Access Facility, Energy Access Ventures, Facility for Energy Inclusion**) to quickly scale impact and get Africa on the low-carbon SDG7 trajectory.



to monetize social & environmental impact of DRE enterprises, such as **Universal Energy Facility, Decentralized Renewable Energy Credits, and Digital Carbon Credits.**



that leverage the co-benefits of the SDG7-SDG13 nexus & **roll out new solutions to unlock climate funding for the DRE sector.**



→ Commissioned by **Shell Foundation & Rockefeller Foundation**

Shell Foundation |



The
**ROCKEFELLER
FOUNDATION**

→ Produced by **Catalyst Off Grid Advisors**



CATALYST

OFF GRID ADVISORS

→ Backup genset research produced in association with the **Schatz Energy Research Center**



**SCHATZ
ENERGY
RESEARCH
CENTER**

→ Inputs from a **Technical Working Group**, comprised of representatives from the following organizations



IKEA Foundation



→ The Report's backup genset research was based on previous work commissioned by the **International Finance Corporation**



IFC | International
Finance Corporation
WORLD BANK GROUP

→ The Report is endorsed by key **DRE industry associations**

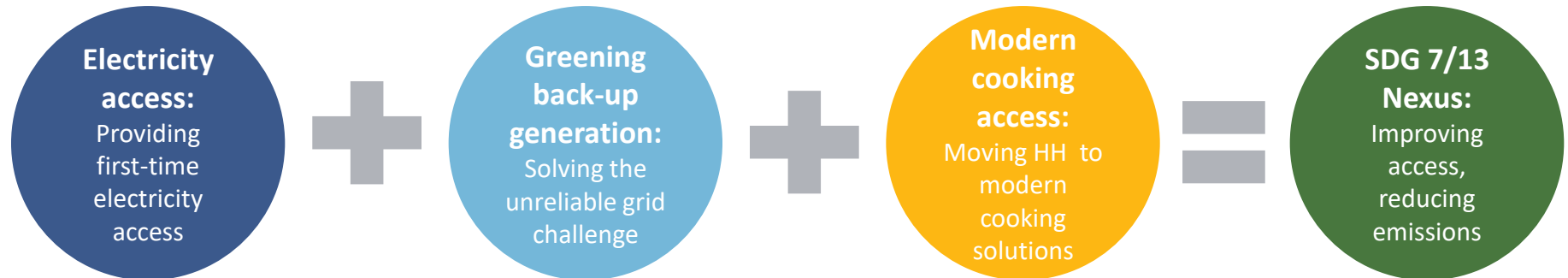


**CLEAN
COOKING
ALLIANCE**



Annex: research methodology & additional analysis

Key questions guide the research agenda



- How to transition households off stopgap solutions and onto modern, affordable, and reliable sources of electricity?
- What are the avoided emissions associated with a low-carbon universal electrification scenario (and in a high demand context)?
- What would it cost to make this a reality?
- How do we get enterprises and households off their dependency on backup fossil-fueled generators (backup genset)?
- What are the emissions from the use of backups, and what would the counterfactual look like?
- How much would it cost to replace backup generators with renewable backup technologies?
- What are the emissions associated with traditional cooking methods?
- What would it look like if a meaningful percentage of users were to graduate to modern cooking solutions?
- What would this cost and what are some of the other key considerations?
- What is the climate impact (measured in GHG emissions) of different universal electrification (and improved cooking) scenarios?
- What is the climate finance opportunity associated with a low-carbon universal electricity access scenario?

Research scope: focused on portions of SDG 7 and 13

Climate Finance Opportunity

Our Coverage

Global



Africa, India, Myanmar only

SDG 7: Access to affordable, reliable, sustainable, and modern energy for all



Off-Grid Solutions for Households



Grid Extension for Households



Cooking access for Households



Electricity Access for Enterprises

SDG 13: Take urgent action to combat climate change and its impacts



Carbon Scenarios



Avoided Emissions

Increase share of renewables



Households (Backup genset)



Enterprises (Backup genset)

Leveraging climate finance for SDG 7/13 nexus



Universal electrification



Improved electricity access



Cooking access

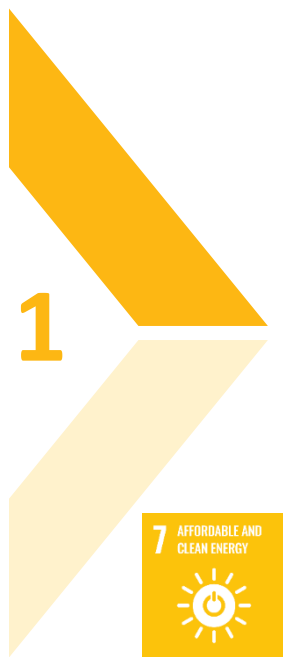


Full alignment



Partial alignment

We built empirical models, with 5 main components



Modeling to derive each technology's **contribution towards achieving SDG 7**, via different scenarios, including 'high demand' universal access



Modeling **indirect** GHG emissions of each electricity **generation & cooking technology**, plus those from **stopgap lighting**



Modeling **direct** GHG emissions from **household and enterprise electrification**, **household cooking**



Using emissions modeling results from **step 2+3** to derive **total emissions** from step 1 scenarios



Modeling the **Climate Finance Opportunity** in Africa, India, Myanmar

We modeled various SDG 7 scenarios and the associated carbon footprint

Core Data (Fixed Assumptions)

Historical access rates, from

Grid/MG/
SHS

Genset

Improved cooking

Stopgap technology use

Lifecycle Assessment

Population growth

Key Levers (Var. Assumptions)

Future technology blend

Standalone solar

Grid

Mini-grids

Carbon intensiveness of connections

Pace of Access
(including high demand & cooking)

Outputs

2030 access rate

Carbon footprint of each scenario

Access contributions, by technology

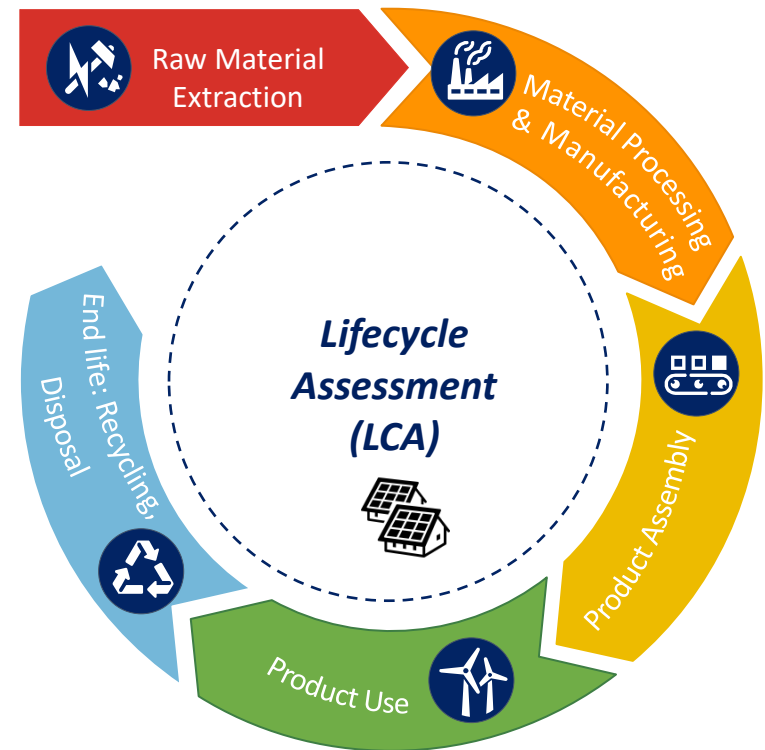
Climate finance opportunity

Levers: can be adjusted to generate a variety of key scenarios

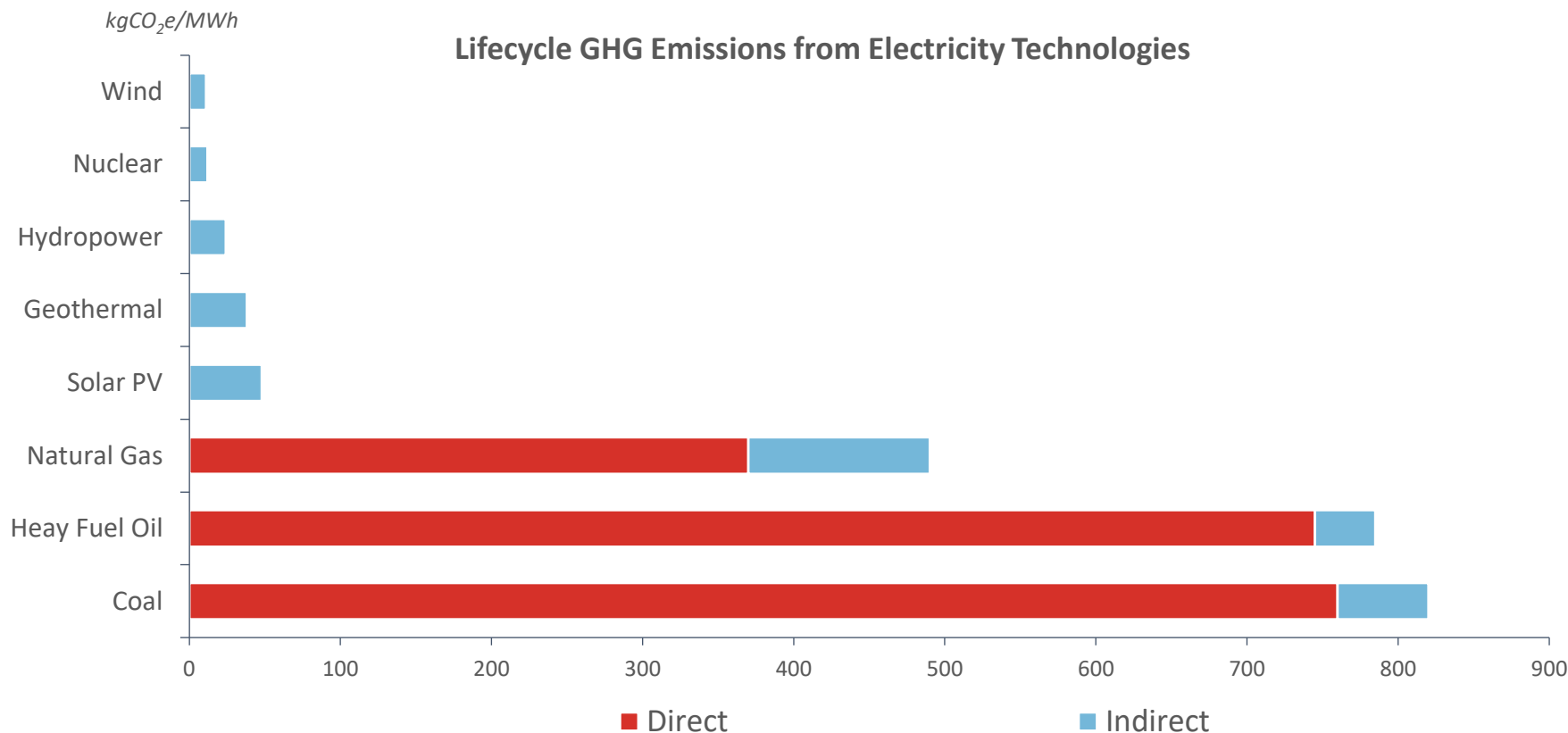
We undertook lifecycle assessment to understand the GHG footprint of each generation technology

Every energy technology has a carbon footprint, even low-carbon technologies

- ‘Cradle to grave’ approach in **lifecycle assessment (LCA)**
 - Assesses environmental impact from extraction (cradle) to end use (grave) of **energy technologies**:
 - Grid-tied generation (e.g., solar, gas)
 - Mini-grids
 - Standalone home systems
 - Stopgap solutions
 - Standalone genset
- LCA of energy technologies to capture **GHG emissions**
 - LCA captures CO₂ associated with the production of all energy technologies
 - No carbon footprint from operation of solar technologies (unlike fossil fuels)



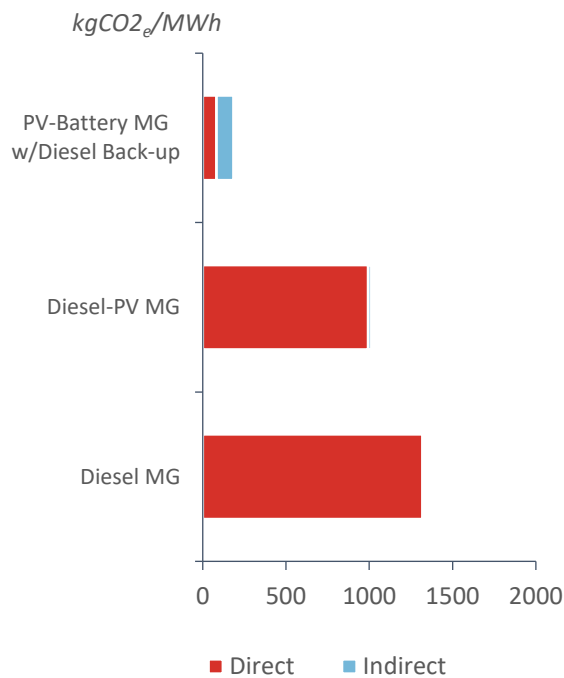
Lifecycle GHG emissions of utility-scale power generation technologies



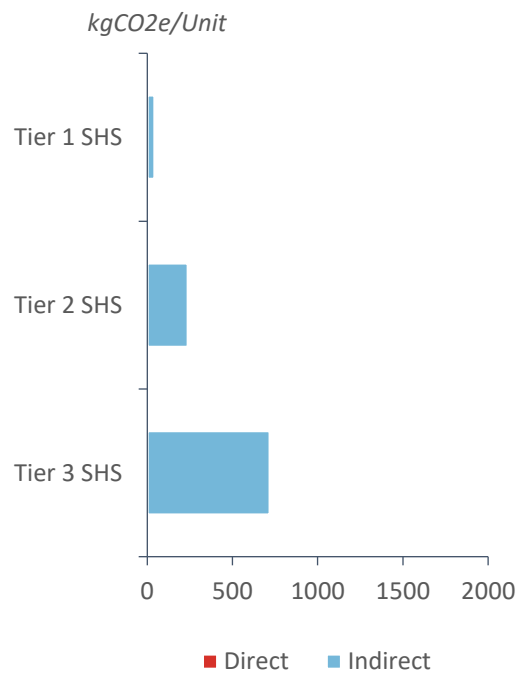
- **Indirect** GHG emissions: generated from extracting, processing and manufacturing activities associated with manufacturing, transporting, and assembling generation technologies
- **Direct** GHG emissions: generated from operation of generation technologies, post-installation

Standalone, mini-grids, and genset: varied units of measure, and significantly different GHG footprints

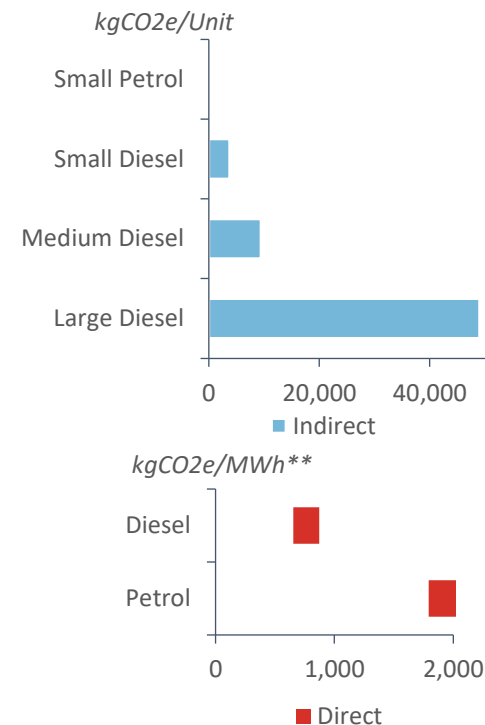
Mini-Grid Systems



Standalone Solar



Genset



- Off-grid and mini-grid technologies offer **lower-emissions access scenario**, but *only if* fossil fuel-based generation sources are minimized
- While the level of service provided by **Tier 1** standalone solar solutions is limited to lighting and basic charging, HH can graduate to larger **Tier 2, 3** systems as needs and purchasing power increase

Emissions from stopgap lighting solutions vary drastically as well...

Modern, reliable electricity technologies often displace massive amounts of CO₂e emissions

1,000 kg CO₂e

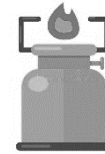
emitted per year on kerosene lanterns, and particularly open wick models

15 kg CO₂e

emitted per year on battery-powered torches, mobile-phone torches, or small solar lanterns



Kerosene
(hurricane lantern)



Kerosene
(open wick)



Candles



Mobile Phone Torch



Flashlight
(dry cell-powered)



Wood

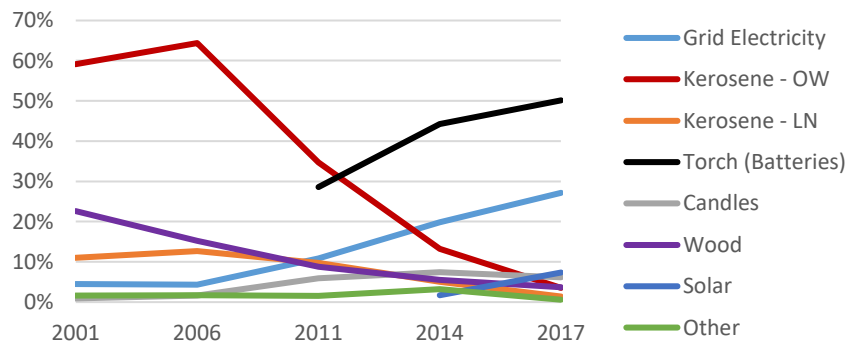
...though our research shows a promising shift away from high polluting stopgap technologies (1/2)



Rwanda

- HHs using kerosene as primary lighting source dropped from 70% to 5% between 2001 and 2017
 - In 2017, kerosene accounted for <10% of stopgap lighting used in the country
 - Torches powered by dry-cell batteries now primary lighting source for 70% of unelectrified HHs

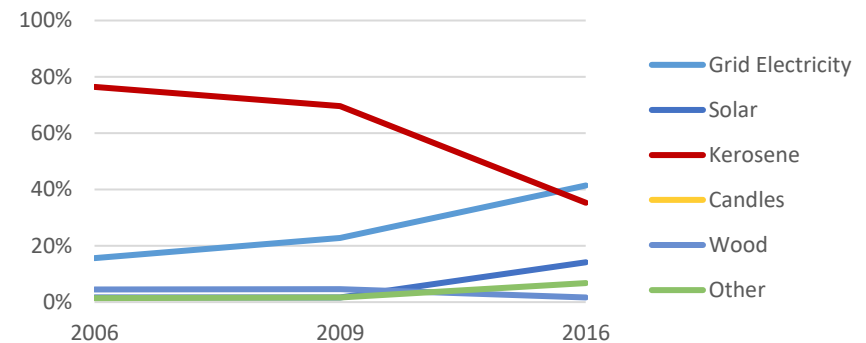
Rwanda - Primary HH Lighting Source



Kenya

- Kenya is a major outlier, with sticky kerosene users who switch mostly to the grid or solar
 - In 2017, kerosene still accounted for nearly half of stopgap lighting used in the country
 - Torches were the primary lighting source for just 5% of households in 2016

Kenya - Primary HH Lighting Source



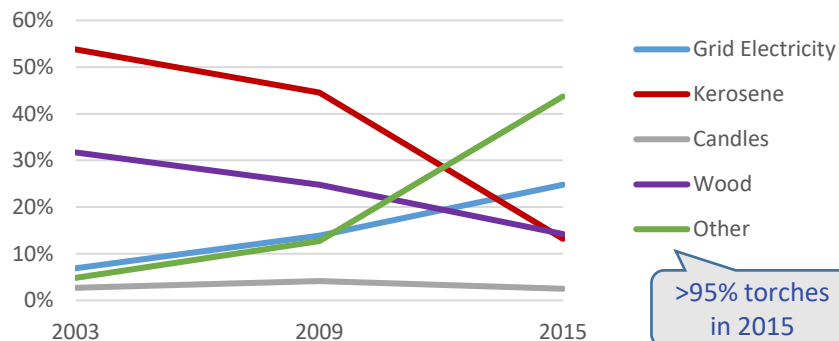
Country-level deep dives show a striking shift in kerosene usage (2/2)



Mozambique

- HHs using kerosene as primary lighting source dropped from 54% to 13% between 2003 and 2015
 - In 2015, kerosene accounted for <20% of stopgap lighting used in the country
 - Torches powered by dry-cell batteries are primary lighting source for >40% of unelectrified HHs

Mozambique - Primary HH Lighting Source



Uganda

- HHs using kerosene as primary lighting source dropped from 85% to 34% over the last decade
 - Torches powered by dry-cell batteries now primary lighting source for >21% of unelectrified HHs
 - In 2017, kerosene did still account for approx. half of stopgap lighting used in the country

Uganda - Primary HH Lighting Source

